

Optimal Execution in Presence of Short-Term Trading

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Abstract

Starting from basic hypotheses on how footprints from hidden orders are interpreted by short-term traders, we derive a fair price model that predicts market impact for non-uniform participation rate schedules. We use this model to derive an optimal execution schedule for a risk-averse trader. The optimal schedule delays front-loading to avoid the information shock of an abrupt start. We also consider optimal strategies with respect to the Volume-Weighted Average Price (VWAP) benchmark. We show that the VWAP-optimized schedule for a large order is similar to the risk-averse one. In an example, we compute the cost of front-loading, and the additional cost of the information shock that results from an aggressive trade start.

Key Words: Hidden order, information leakage, market impact, trading cost, optimal trade execution.

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Introduction

In portfolio management, implementation shortfall is the portfolio loss resulting from the difference between the realized price of a trade and the market price at the time of the decision. The aggregate cost society pays for active investing has been estimated to represent 67bps annually for the US stock market (French, 2008); at a 2011 market valuation this would represent \$105bn/year. Part of this cost is realized through the implementation shortfall of institutional trades. However, fund performance could be ameliorated by the design of execution strategies that minimize implementation shortfall.

In this paper, we consider the optimal execution of a single large portfolio transaction, realized through a hidden order that must be split into smaller slices and executed incrementally over time. We consider the case where the execution proceeds without interruption, at a sufficient rate and for a long enough time to cause observable effects on the order flow. A participation rate schedule consists in a sequence of participation rates, each of them applied in a particular interval. The optimization problem is to find the participation rate schedule that minimizes the risk-adjusted cost while completing the trade in a given maximum amount of time.

One must first specify a model for market impact to optimize the risk-adjusted cost. Different impact models lead to different optimal execution solutions (Kyle, 1985), (Bertismas and Lo, 1998), (Almgren and Chriss, 2000), (Almgren *et al.*, 2005), (Obizhaeva and Wang, 2013). From the hypotheses established in our theory, we will obtain a market impact function that, in each transactional interval, *depends non-linearly* on both the total number of shares traded from the start of the trade until the observed time, and the participation rate at that interval. Additionally, impact at each step will be *non-stationary* because of its dependency on the time elapsed from the start of the trade.

In comparison, Almgren and Chriss assumed that the market prices are driven by an arithmetic random walk overlaid with a *stationary* market impact process. Impact is proposed to be the linear sum of permanent and temporary components, where the permanent impact *depends linearly* on the number of traded shares and the temporary impact is a *linear function* of the trading velocity. They find that, with risk-aversion, it is optimal to trade fastest at the beginning and slow down as the trade progresses (front-loaded execution) according to a hyperbolic sine function.

This type of front-loaded participation rate profile is widely used by industry participants to reduce risk, yet it is also recognized that front-loading executions increases impact early in the trade resulting in higher trading costs. A related concern is that liquidity exhaustion

or increased signaling risk could also lead to a higher variance in trade results (Hora, 2006), defeating the main purpose of front-loading. The non-linearity of temporary impact in the trading velocity has been addressed previously in (Almgren, 2003), (Almgren *et al.*, 2005); the optimization method has also been adjusted for *non-linear* phenomenological models of temporary impact (Loeb, 1983; Lillo *et al.*, 2003). However, these studies share the assumption that the effect of trading on price is *stationary* for a uniform participation schedule; i.e., temporary impact remains constant for the duration of the trade (explicitly independent of time). Instead, practitioners find that total impact is a concave function of the amount of time that a uniform participation algorithm has been engaged.

In previous work (Farmer *et al.*, 2011) (FGLW) proposed a *non-stationary* model for market impact *for constant participation rate*. In this theory, the basic assumption is that traders are able to detect the existence of a hidden order by observing an imbalance in the order flow, caused by the execution algorithm. Temporary impact represents expectations of further activity from this algorithm. Since a hidden order that has been observed for a longer period is more likely to continue, temporary impact grows. The continuation probability depends on the distribution of hidden order sizes. If one assumes that the hidden order size has a Pareto distribution with a tail exponent of 1.5, the model predicts that temporary and permanent impact of a constant-participation algorithm both grow as a square root of time. Integrating over time to compute the average price, it follows that total impact is also a square root of the trade size. This result is in agreement with phenomenological models including the Barra model (Torre, 1997). See also, (Chan and Lakonishok, 1993), (Chan and Lakonishok, 1995), (Almgren *et al.*, 2005), (Bouchaud *et al.*, 2008), (Moro *et al.*, 2009).

In the FGLW approach and in this paper, the impact of an individual transaction depends on the prior activity of the hidden order. Other models have considered impact to be the sum of independent transactional impacts with a decay kernel; for example: (Bouchaud *et al.*, 2004), (Alfonsi *et al.*, 2008), (Gatheral, 2010), (Obizhaeva and Wang, 2013). In (Huberman and Stanzl, 2004), it is argued that permanent impact must be linear in order to prevent the existence of systematic profits from price manipulation. However, this argument assumes that the permanent impact functions are the same for the buy and sell periods of a round-trip trade. Permanent impact is the expected information content of a trade; our framework deals with trades that are large enough to be detectable so market observers, who see a large buy followed by a large sell, cannot be expected to believe that both trades are equally informed.

This article is organized as follows. In the first section, we extend FGLW's theory to allow for a variable participation rate. The

generalized theory has additional hypotheses and is self-contained. In the second and third sections, we use numerical methods to find optimal solutions that minimize the risk-adjusted cost and optimize the performance to the VWAP benchmark. Section fourth discusses the implications our model has for institutional trading desks. An Appendix contains detailed derivations of some formulae.

1. Short-Term Trading and Hidden Orders

Our paper addresses the situation of a large institutional trade of size X , that is executed over time through a sequence of smaller transactions. For simplicity, we will consider a single institutional trade executing in a market where prices are driven by an arithmetic random walk. The trade is executed according to an execution schedule with participation rates $\pi(s)$, which represents the probability that a market transaction s belongs to the institutional trade.

For large enough trades and high enough participation rates, it seems reasonable to assume that market observers will be able to detect the presence of the institutional trade through statistical means, with a degree of confidence that increases over time. For example, one might observe a significant imbalance in order flow, or short-term changes in the security's price relative to some index.

A class of short-term trading strategies arises with the aim to profit from information about hidden orders. If short-term traders believe that a hidden order will continue, they will expect further impact and take a position on the same side. On the other hand, if they believe the hidden order will stop, they will expect reversion and take an opposite position. The competition between short-term traders sets the fair value of market impact.

Our next step is to capture order flow inputs and other concepts as hypotheses for a theory of impact. In Section 1.2, we will explicitly obtain the mathematical function that incorporates all the hypotheses and models impact. The equation of price dynamics that incorporates impact will be exposed in Section 1.3.

1.1 Hypotheses

- **(H. 1) Hidden Order Detection.**

A hidden order with participation rate π can be detected at or after the end of an interval of $\tau(\pi) = 1/\pi^2$ market transactions.

Justification:

We will consider an anonymous marketplace where order arrivals can be classified as either buyer-initiated (+sign) or seller-initiated (- sign).

We assume that in absence of a hidden order, both signs are equally likely. A hidden order originates only buy or only sell orders. We will call the difference between buyer and seller-initiated transactions in an observation interval, *the order flow imbalance*.

We suppose that all market transactions are for the same number of shares n . Market participants, who detect an order flow imbalance, wish to determine whether it originates from a hidden order executing with a participation rate π , where $0 < \pi < 1$, or as a random occurrence from equiprobable buy and sell transactions. Given such a hidden order, the expected imbalance will be $\tau\pi$ for a market observation period of τ transactions. If there were no hidden order, the expected imbalance would be null with a standard deviation of $\sqrt{\tau}$ (for $\tau \gg 1$). Therefore, the hidden order becomes detectable with $t - \text{stat} \geq 1$ when $\tau\pi \geq \sqrt{\tau} \Leftrightarrow \tau \geq 1/\pi^2$. Otherwise, if $\tau\pi < \sqrt{\tau}$, the imbalance may be interpreted as market noise in a situation when no hidden order is transacting.

For simplicity, we will discretize the space of execution schedules by considering piecewise flat schedules that maintain a constant participation rate π in *the minimal interval of detection* $\tau \equiv 1/\pi^2$. In practice, there is no particular reason why an institutional trader should choose to change participation rates after precisely $1/\pi^2$ market transactions. However, we will show in Section 4.2 that the qualitative features of these optimal solutions remain the same if one chooses coarser discretizations.

We will call intervals satisfying the detection criterion “*detectable intervals*”, and the index k will represent the k^{th} *detectable interval*. We consider as *an execution schedule* a finite set of detectable intervals, where the institution maintains a constant participation rate within each interval. The choice of a trading trajectory by the institution affects how an order becomes detectable; a higher participation rate will cause detection to occur sooner. If an institution is executing a hidden order with a participation rate of π_k , it should expect to become detectable after $\tau_k = \pi_k^{-2}$ market transactions. During this k – *interval*, the institution expects to execute a total of π_k^{-1} partial fills. For example, a hidden order executing with a uniform participation rate of $\pi = 0.1$ is detectable after a segment of 100 market transactions. In this interval, it will have accumulated 10 partial fills. The average trade size in the US, at the time of this writing, is approximately $n = 250$ shares; therefore, such a hidden order becomes detectable after 2500 filled shares.

The order will be traded in a finite amount of time; then, a finite number of detectable steps will take place. This number can be represented by an integer that we will call $[N^*] \stackrel{\text{def}}{=}$

$IntegerPart(N^*)$, where $N^* = [N^*] + q$, $0 \leq q < 1$. In other words, $[N^*]$ represents the last *detectable* interval. A hidden order will generally not stop precisely at the end of a detectable interval $[N^*]$. A remaining time of trading τ_r could exist that is not sufficient to complete another detectable interval of $1/\pi^2$ market transactions. We set $\tau_r = \tau_r(q, \pi_f) = q \pi_f^{-2}$, where π_f is the most recently observed rate. Therefore, an execution schedule will extend over a total number of intervals N^* , which need not be an integer. The number N^* will be determined by the trade size X and the transactional time horizon T in a constrained optimization problem that we will explain in Section 2.

In this paper, we will not concern ourselves with the relation between the transactional time interval $\tau(\pi)$ and actual clock time. Each interval has a different length as the rate varies and it is dimensionless. For the purposes of this paper, the number of market transactions is the natural scale to measure time.

We define the *non-random* variables of the process of executing a hidden order to be $\{\pi_1, \pi_2, \dots, \pi_{[N^*]}, \pi_f, N^*\}$. That means the trade will be executed by the institution with the participation rates schedule $\vec{\pi} = \{\pi_1, \pi_2, \dots, \pi_{[N^*]}, \pi_f\}$ and size N^* . The number of variables will be reduced once we set the constraints of the optimization problem for the total cost of the trade.

Figure 1 (see page 40) illustrates the scenario we present in this paper. The execution schedule is divided in intervals, each with a uniform participation rate. Time flows from left to right. The number of market transactions and shares filled are shown together with the changes of participation rate.

The process of price dynamics will be modeled as a stochastic process; therefore, we will need to define a distribution function for market prices at any time or the type of process itself.

- **(H. 2) Process of Price Dynamics.**

Before introducing the second hypothesis that describes the evolution of prices, we will explain some concepts and definitions.

H.2.1 Prices and Impacts

As explained before, our model is designed for the case where a market is driven, in part, by an institution that executes a hidden order. Observers can expect two different situations at interval k : the institution either stops or continues the execution of the order. As a result, two kinds of market prices and two kinds of impacts arise.

By **reversion price**, S_k , we will mean the future price per share conditioned on observers determining that the hidden order ended after completing the interval k . The reversion price accounts for the **permanent impact** from prior intervals as:

$$S_k \supset S_j + \text{Permanent Impact}(\text{from } j \text{ to } k), \forall k > j. \quad (h.2.1)$$

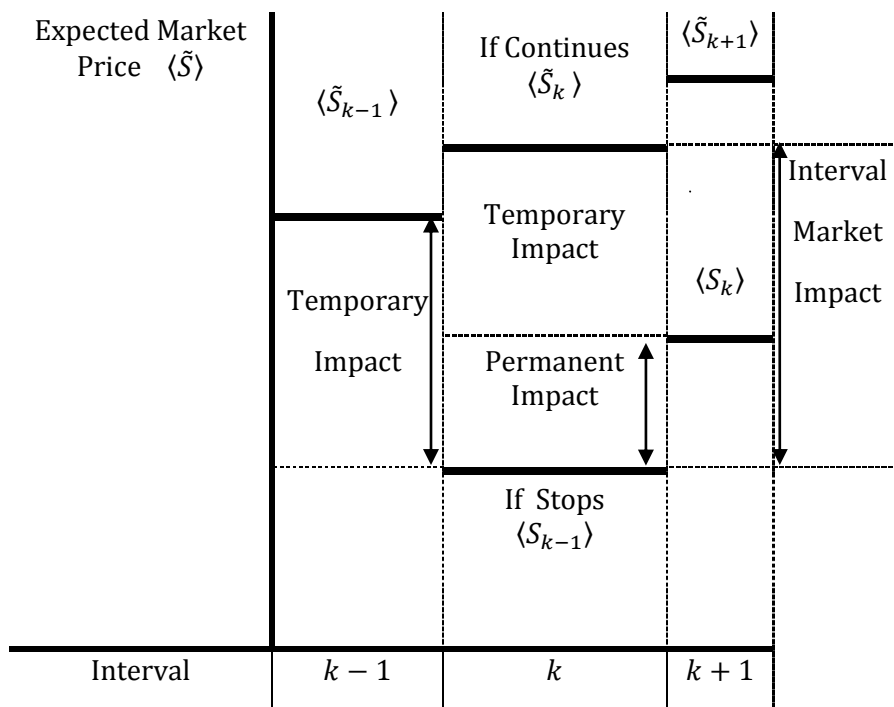
We will denote by $\tilde{S}_k$ **the execution price** per share in the k -interval. It is the market price conditioned on that the hidden order continues through interval k . The execution price contains the sum of the **temporary impact** and the reversion price at the same interval. However, relative to an earlier interval, it accounts for both permanent and temporary impacts what we call simply **Impact**. That is:

$$\tilde{S}_k \supset S_k + \text{Temporary Impact}(k), \quad (h.2.2)$$

$$\tilde{S}_k \supset S_j + \text{Impact}(\text{from } j \text{ to } k), \forall k > j. \quad (h.2.3)$$

The **reversion price** is hypothetical if the institution does not stop at the end of the interval. **Figure 2** shows these concepts graphically.

Figure 2. Definitions of Prices and Impacts. The expected market price at k is $\langle \tilde{S}_k \rangle$ if the institution continues after $(k-1)$; if it stops, the market price is the reverted price $\langle S_{k-1} \rangle$. Temporary Impact at $(k-1)$ is the difference $\langle \tilde{S}_{k-1} \rangle - \langle S_{k-1} \rangle$, and Permanent Impact at k is the difference $\langle S_k \rangle - \langle S_{k-1} \rangle$. Interval-Market Impact at k is $\langle \tilde{S}_k \rangle - \langle S_{k-1} \rangle$, and it is the sum of Permanent and Temporary Impacts at k .



We could construct the dynamics of prices from any of the expressions given above. However, we are interested in whether a hidden order that has completed $k - 1$ intervals will continue or stop. Following the definitions of prices, if the institution stops, the price will reach the reversion price S_{k-1} ; if it continues, it will reach the market execution price $\tilde{S}_k$.

The difference $\tilde{S}_k - S_{k-1}$, will comprise two conceptually different contributions. One will be predictable and known as *market impact*, or simply *Impact* as explained in (h.2.3). The other one is the unpredictable *noise term*.

H.2.2 Variables and functions of the dynamic process

Following with the ideas exposed above, we will construct the prices of the asset at each step k as functions depending on the variables of the execution schedule, $\{\pi_i\}_{i=1}^k$, and on the independent random variables of Gaussian white noise, $\{\zeta_i\}_{i=1}^k$.²

We will define $Impact_k(\{\pi_i\}_{i=1}^k)$ as the *interval–market impact function* at k , depending solely on the variables of the execution schedule. This term represents the change of price derived by the law of supply and demand in the interval τ_k , principally driven by the institution and the observers who are able to detect it and try to make profits. We will assume in Hypotheses (H.3) and (H.5) that profits are zero on average, and show that this defines, in part, the shape of the impact function. The expression for $Impact_k$ will be qualitatively proposed in Hypothesis (H.4) and explicitly derived in Section 1.2.

On the other hand, the stochastic aspect of prices will be modeled by the *noise term*, which will consequently be proportional to the volatility of the asset σ and depending on the random variables of noise. In this paper, we will consider a *short-term trading in non-highly volatile markets*; therefore, the volatility σ can be considered as a constant independent of the prices. Mathematically, *the noise term in the interval τ_k* will be represented by a small change in a variable z_k , which follows a Weiner process of *pure Brownian motion or diffusion* in the short-term trading. Explicitly, this means $\Delta z_k = \sigma \tau_k^{1/2} \zeta_k$. Additionally, if the function $G = G(\{\zeta_k\}_{k=1}^{N^*})$ is defined as the Gaussian distribution on the independent random variables of white noise, it corresponds to a Gaussian mean

² In trading, noise means all factors that change the price in an unpredictable manner and, consequently, should be studied statistically through their correlations and cumulants related to a distribution function. *Gaussian white noise* is defined by a Gaussian process with zero mean and delta correlation. It has been chosen in this paper for its mathematical simplicity and is widely used in models of trading.

$\langle \zeta_k \rangle_G = 0$, Markov correlations $\langle \zeta_k \zeta_j \rangle_G = \delta_{kj}$, and standard deviation $\sqrt{\langle (\Delta z_k)^2 \rangle_G} = \sigma \tau_k^{1/2}$.

Until here, we have collected all the elements necessary for the introduction of our next hypothesis.

H.2.3 Evolution of prices

Inspired by the literature in optimal trading execution, we will consider the process of price dynamics as a discrete-time Langevin process³. The intervals between steps k and $k - 1$, will be determined by $\tau_k = \pi_k^{-2}$ market transactions.

We will suppose the drift term $Impact_k(\{\pi_i\}_{i=1}^k)$ and the volatility σ independent of the prices. This kind of discrete-time process is usually called *arithmetic random walk*. Summarizing:

The process of trading drives market prices as:

$$\begin{aligned} \tilde{S}_k(\{\pi_i\}_{i=1}^k, \{\zeta_i\}_{i=1}^k) \\ = S_{k-1}(\{\pi_i\}_{i=1}^{k-1}, \{\zeta_i\}_{i=1}^{k-1}) + Impact_k(\{\pi_i\}_{i=1}^k) \\ + \sigma \tau_k^{\frac{1}{2}} \zeta_k, \end{aligned} \quad (h.2.4)$$

with the definitions given in H.2.1 and H.2.2.

In hypothesis (h.2.4), we are considering the variables $\vec{\pi}$ non-random and independent of the variables $\{\zeta_k\}_{k=1}^{N^*}$, or the same: $\vec{\pi} \neq \vec{\pi}(\{\zeta_k\}_{k=1}^{N^*})$. That can be valid in non-highly volatile markets for algorithms that are designed to track a specific execution schedule. Then, when evaluating the expected value on equation (h.2.4), the impact term will remain the same and the noise term will vanish. The expected change in price during the interval τ_k will depend on the trading schedule $\{\pi_i\}_{i=1}^k$ through the particular shape of the impact function as:

$$\langle \tilde{S}_k - S_{k-1} \rangle_G = Impact_k(\{\pi_i\}_{i=1}^k). \quad (h.2.5)$$

The expectation is over the Gaussian function $G(\{\zeta_i\}_{i=1}^k)$, with fixed variables $\{\pi_1, \pi_2, \dots, \pi_k\}$.

Comments:

³ See for example (Cobb, 1998).

It follows from (h. 2.4)⁴ that:

On short-term trading in non-highly volatile markets, the distribution function of the market prices is a Gaussian distribution of a Wiener process with constant drift term. Therefore, market prices are determined uniquely by their expectation values and standard deviations (first and second cumulants). The expectation value is related to the total impact of the trade, while the standard deviation is proportional to the square root of the elapsed transactional time.⁵

- **(H. 3) Breakeven.**

The expected reversion price following a trade that completed after k intervals is equal to its weighted average execution price:

$$\langle S_k(\{\pi_i\}_{i=1}^k) \rangle_G = \frac{\sum_{i=1}^k \frac{1}{\pi_i} \langle \tilde{S}_i \rangle_G}{\xi_k}, k > 1. \quad (h. 3)$$

Here, $\xi_k = \sum_{i=1}^k \frac{1}{\pi_i}$ is the number of institutional transactions to step k , and π_i^{-1} is the number of institutional transactions in each step i .

Justification:

⁴ Because a Wiener process postulates the *independency* of all its Gaussian variables $\{\zeta_k\}_{k=1}^{N^*}$ then *their sum is also Gaussian*. In Section 1.3, we will derive equation (7) that gives the expression for the market execution price at an interval as *the sum of the variables of noise* and the expectation value of the execution price, which is the total impact accumulated since the start of the trade. Therefore, we conclude that the market price is a Gaussian random variable. In this paper, we are interested in the expression of the expected price in terms of the market impact and the execution schedule and we will never use the explicit form of the Gaussian distribution $G = G(\tilde{S}_k)$ on the price random variable. However, we note that the *arithmetic process* makes $G(\tilde{S}_k, S_0) = \text{Normal}(\langle \tilde{S}_k \rangle_G, \sigma \sqrt{\sum_{i=1}^k \tau_i})$ in comparison with a *geometric process* $G(\ln(S(t)/S_0)) = \text{Normal}(\langle \ln(S(t)/S(t_0)) \rangle_G, \sigma \sqrt{(t - t_0)})$, where t is the transactional time.

⁵ We know that this result is controversial in general cases, where stock prices could be strenuously fluctuating and follow other process as complicated as fractal ones. In those cases, cumulants of order greater than 2 are required. See for example (Peters, 1994), (Bouchaud and Sornette, 1994), (Ghashghaie *et al*, 1996), (Mandelbrot, 1997). However, for non-highly volatile markets we are no so far from reality if we work with cumulants to order two and in a diffusive regime (or pure Brownian motion) for the short term trading, where the variance is proportional to the time.

The breakeven hypothesis states that the value of the shares bought (or sold), after reversion, is the same as the average price paid (or received) in the execution. For example, suppose a portfolio manager places a buy order when the price is \$20. As the hidden order executes, the price rises to a peak of \$21, with an average price of \$20.7. After the trade ends, price reverts part way and closes at \$20.7, leaving no profit after reversion. In this example, we assumed the noise was zero for clarity. Actually, some trades will be profitable after reversion and others unprofitable, Hypothesis (h.3) states that breakeven will hold when averaged over the Gaussian noise distribution. We regard the breakeven hypothesis primarily as an empirical observation, see (Bershova and Rakhlin, 2013) and below. However, it raises an interesting question: if the security price were to revert to \$20.7 in the above example, why would the manager continue to buy as it rises above this level? The main reason is that noise term in equation (h.2.4) creates uncertainty in the execution price, which represents risk to the portfolio manager proportional to the elapsed transactional time. To manage this risk, portfolio managers expect trades to be completed in a fixed amount of time. Should one try to rush the trade to complete the order before price reaches \$20.7, market impact will increase faster making the realized execution price even higher. Given the constraints on the total size of the order and the total amount of time available to trade, reversion is an inevitable consequence of the removal of order flow expectations after a hidden order has stopped.

The breakeven hypothesis contrasts with Kyle's informed trader model. In Kyle, the insider knows the final price and, therefore, would not logically buy above this level; impact is linear, and there is no reversion. As a result, Market Makers in Kyle's model systematically lose money to hidden orders and generate counterbalancing profits at the expense of the uninformed traders. Kyle's model predicts a net transfer of wealth from uninformed traders to informed traders. We instead theorize with the breakeven hypothesis that hidden order expected profits are null; therefore, our model does not require noise traders to lose money systematically. Both models are idealizations of situations that arise in the real world⁶.

We test this hypothesis using datasets provided by Alpha Vision clients. The data contains 86253 trades larger than 1% of daily volume from 94 portfolio managers in European and US cash equities, over various time windows from May 2008 through November 2012. To confirm that the equality (h.3) is fulfilled, we

⁶ Reality lies somewhere in between these two idealizations. For example, portfolio managers need at times to execute liquidity trades in spite of an expected loss. Vice-verse, it may be possible in some circumstances to identify informed traders where the portfolio manager makes a market-to-market profit using techniques from data mining.

first need to adapt this equation to a real market situation. Next, we describe how we proceed.

1. *k – intervals:*

It is common for various portfolio managers at a given asset manager to place orders in the same security, in which case the desk usually aggregates orders into one and executes the merged trade as a unit. Therefore, we will merge overlapping orders from a given firm. In addition, we will also merge orders that are separated by no more than 60 market minutes, where we count the time from the market close to the next-day open as zero market minutes. We will bin merged trades by their size X , which is defined as the total number of shares traded. We will suppose that for each interval $k > 1$, there exists a hidden order that completes its entire order size X in $[N^*(X)] = k$ intervals. Therefore, and just for the purpose of the verification of breakeven, we will use order sizes X in place of k –intervals.

We will call the right side of equation (h.3) “the averaged capture”, and the left “the reversion price”.

2. *Averaged Capture and Shortfall:*

We calculate the average execution price to fill the entire merged trade or *average capture* as a volume-weighted average fill price of all partial fills.

For each merged trade, we consider the midpoint price at the start of the trade that we call *start price* or *AP*. We take

$$AP = \frac{Bid + Ask}{2},$$

at the starting point of the trade. The relative difference between the average capture and the start price is the *shortfall*. In basis points, this is:

$$shortfall \stackrel{\text{def}}{=} \left(\frac{average\ capture - AP}{AP} \right) \times 10^4 \times sign,$$

where *sign* of the trade is (+1) for a buy and (−1) for a sell.

3. *Reversion Price and Return:*

The reversion price is the price relaxation after a trade has ended. It is difficult to measure because portfolio managers often add positions in subsequent days. We remove the effect of these follow-on trades by subtracting their estimated impact, using an empirical impact model that we explain below in “First- Interval Impact”. We

will call the adjusted amount the *free price* because it represents the value of the security free of further activity.

The reversion price is the free price when the market determines that the hidden order has stopped. It is difficult to establish when an order has ended. One of the reasons is that the market does not know whether the institution is making a pause or stopping. The data show that very large trades that require several days to execute can take two to five market days to complete reversion. Generally, a full day is sufficient to observe complete reversion. Here we will consider the following options for the reversion price: 1) the price at the market close on the day the trade was finished or *close price* "T", 2) the price on the close of the next day "T+1", and 3) the price on the close after two days "T+2".

Our purpose is to compare the shortfall to the return,

$$return = \left(\frac{reversion\ price - AP}{AP} \right) \times 10^4 \times sign.$$

Below we show **Figure 3** that compares the shortfall with the return at the close, at the close after one day and at the close after two days, as a function of the trade size. We are using a logarithmic scale for a better view.

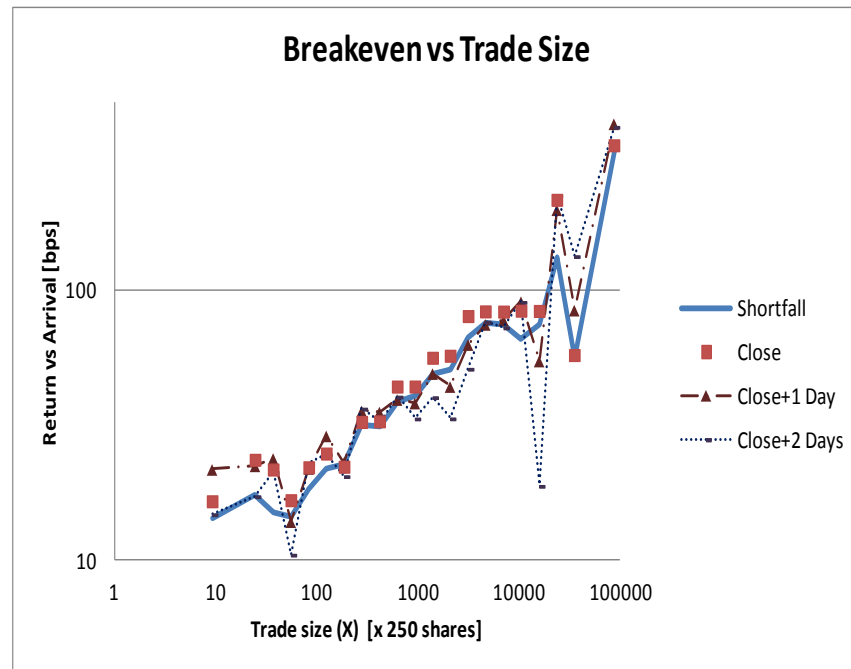


Figure 3. Breakeven is verified by comparing the implementation shortfall (thick line) and the return, with trades binned by trade size X . On the close, we observe some unrealized profits (reversion is not complete) in the economically important range $X \in [200, 10000]$. By Close+1 Day, the difference is insignificant with $t\text{-stat} < 2$ for 90% of the bins shown on the chart.

- (H. 4) Temporary Impact.

H.4.1 First interval impact

The impact in the first interval is equal to the product of a scaling factor, which depends only on the volatility σ , and an exponent $\gamma = -0.2$ of the participation rate in the interval:

$$\langle \tilde{S}_1 \rangle_G - S_0 = \hat{\mu}(\sigma) \pi_1^\gamma. \quad (h.4.1)$$

Justification:

The first interval is the elapsed time before a hidden order can be detected. We adopt a phenomenological approach to model the first-interval impact. We use a proprietary dataset of 43095 constant-participation trading segments corresponding to market orders executed between August 2009 and October 2012. Each *segment* in this dataset is an algorithmic trade starting from order arrival, and executed to completion with a uniform participation rate objective. A segment typically contains several detectable intervals, j , and the algorithmic trading tool offers a choice of three participation rates: 0.1, 0.2, or 0.3.

To limit selection bias, we drop segments that comprise a block execution larger than 50% of the total traded quantity. We also exclude 140 outliers from the small/micro-cap trades representing the top/bottom 1% by shortfall. Additionally, we discard all segments from traders that did not completely fill at least 60% of their orders.

To test the hypothesis (h.4.1), we build a phenomenological model that explains the implementation shortfall of segments of size jX by using a nonlinear regression. Following previous definitions, $S_0 \times \text{shortfall} = \left(\frac{\sum_{i=1}^j \langle \tilde{S}_i \rangle_G}{j} - S_0 \right)$ for this segment that represents the *average impact of a detectable first interval*. We then apply this model to the particular case where X is the number of shares traded in the first detectable interval.

We control for the effect of adverse selection through a variable *ratedev* that measures the difference between intended participation rate and the realized participation rate. Furthermore, we control for market effects through the contemporaneous return of the exchange-traded fund SPY that tracks the S&P500 index. Specifically, *spy* is the product of stock-specific beta and the movement in bps in the ETF for the S&P 500, between the order

submit time or 9:30 for pre-market orders and the order close time, signed by the side of the trade. Additionally, *spread* is the difference in bps between the ask and the bid at the time the order is submitted, or at the market open for pre-market orders. Incorporating all these components, we model the implementation shortfall (in basis points) as:

$$\text{shortfall} = b_1 \sigma^{b_3} \pi^{b_2} X^{b_4} (1 - b_5 \text{ratedev}) + b_6 \text{spread} + b_7 \text{spy}. \quad (h. 4.2)$$

Values for *shortfall* and the size X are obtained from the data and proceeding as in hypothesis (H. 3) point 2. The participation rate π takes one of the values 0.1, 0.2, or 0.3. The parameters $\{b_i\}_{i=1}^7$ are determined through the fitting of the data by the method of the nonlinear least squares using Stata Release 12 (StataCorp., 2011), and the results are given in **Table I**.

	Large/Mid	Small/Micro
b1	0.28 (0.11)	0.19 (0.15)
b2	.28 (0.09)	.4 (0.18)
b3	.89 (0.1)	.87 (0.15)
b4	.59 (0.03)	.35 (0.06)
b5	-.03 (.003)	-.02 (.002)
b6	0.25 (0.04)	0.26 (0.05)
b7	0.45 (0.01)	0.37 (0.02)
R2	0.3433	0.2792
Obs	38,824	4,271

Table I. Results for market orders executed from August 1, 2009 to October 2, 2012. The parameters $\{b_i\}_{i=1}^7$ are determined through the fitting of the data to the equation $\text{shortfall} = b_1 \sigma^{b_3} \pi^{b_2} X^{b_4} (1 - b_5 \text{ratedev}) + b_6 \text{spread} + b_7 \text{spy}$. We estimate the model by the nonlinear least-squares using Stata Release 12. The numbers in parenthesis are robust standard errors provided in Stata 12 that account for heteroskedasticity as suggested by (Davidson and MacKinnon, 1993 and 2004). Values depend of the executed trading size: Large/Mid and Small/Micro. **R2** is the coefficient of determination. **Obs** is the number of observations.

The results are consistent with a speed exponent $b_2 = 0.3$ and trade size exponent $b_4 = 0.5$ ⁷.

We apply these results to the impact of the first interval for any trade that executes $\xi_1 = X$ shares with $\pi_1 = \pi$ during the first interval. Using $\xi_1 = \pi_1^{-1}$ and dropping the spread term⁸, we write (h. 4.2) as:

$$\langle \tilde{S}_1 \rangle_G - S_0 = \hat{\mu}(\sigma) \pi_1^\gamma.$$

Here, $\gamma \stackrel{\text{def}}{=} b_2 - b_4 = -0.2$, and $\hat{\mu}(\sigma) \stackrel{\text{def}}{=} b_1 \sigma^{b_3} S_0$. The terms *ratedev* and *spy* are part of the noise term in (h. 2.4).

H.4.2 Impact after the first interval:

The impact in an interval $k \geq 2$, $\langle \tilde{S}_k - S_{k-1} \rangle_G$, is a function only of the current participation rate π_k and the total number of shares filled through interval k , ξ_k .

Justification:

This hypothesis generalizes the intuition in the FGLW theory that temporary impact represents the market participants' expectations of additional impact from the remainder of the hidden order, averaged over all possible order sizes X . This expected additional impact depends on the number of shares remaining and the speed with which they will be traded. For the shares remaining, it is reasonable to assume that the number of shares X is always drawn from the same distribution of trade sizes, $p(X)$, regardless of the trading schedule – so the distribution will only be conditioned by ξ_k , as $p(X|\xi_k)$. Likewise, we make the simplifying assumption that the participation rate observed in the most recent detectable interval is the best estimator of future participation rates. If we take these two points, it follows that temporary impact should only be a function of the current participation rate and ξ_k . Mathematically, we express these concepts as:

$$\begin{aligned} \text{Temporary Impact}(k) &\stackrel{\text{def}}{=} \int f(\pi_k, X - \xi_k) p(X|\xi_k) dX \\ &\stackrel{\text{def}}{=} \langle f \rangle_{p(X)}(\pi_k, \xi_k). \end{aligned}$$

⁷ Visual inspection of Figure 3 suggests an exponent of 0.33 for the absolute size as a number of shares; however, this does not consider the mutual dependency between the trade size and a stock's average daily volume: large trades are more likely to be in high ADV stocks. The exponent b_4 pertains to size relative to daily volume.

⁸ The spread can be considered as an additive constant to the total cost, together with commissions. We do not regard spread costs as part of the optimization problem in this paper.

Hypothesis **H.4.2** relates to the total impact in interval k , which is the sum of temporary and permanent impacts. The latter is the change in expectations of the information content in the trade. This suggests that incremental permanent impact should depend only on π_k (the number of shares acquired in this interval is $n_k = 1/\pi_k$) and the total number of shares ξ_k .

- **(H.5) Efficient Pricing for Constant Participation Rate Schedules.**

For a hidden order executing at a constant participation rate, let $P(k+1|k)$ denote the probability at the end of the k 'th detectable interval that the hidden order will continue for at least one more detectable interval. The probability that the hidden order stops at $k \geq 1$ is $p(k = [N^*]) \stackrel{\text{def}}{=} 1 - P(k+1|k)$.

The expected profit of a position acquired at the average price in interval $k \geq 1$ is equal to zero at the end of the interval:

$$P(k+1|k)(\langle \tilde{S}_{k+1} \rangle_G - \langle \tilde{S}_k \rangle_G) + (1 - P(k+1|k))(\langle S_k \rangle_G - \langle \tilde{S}_k \rangle_G) = 0.$$

Justification:

Market makers and other short-term trading programs supply liquidity conditionally on order flow. In presence of a hidden order, such a program will accrue a position at the interval average price $\langle \tilde{S}_k \rangle_G$. When the presence of a hidden order is detected at the end of the interval k , the market maker estimates her future profit considering the weighted average of two outcomes. One outcome is $(\langle \tilde{S}_{k+1} \rangle_G - \langle \tilde{S}_k \rangle_G)$, where the hidden order continues for another detectable interval $(k+1)$. The other one is $(\langle S_k \rangle_G - \langle \tilde{S}_k \rangle_G)$, where it stops after completing k and the price will revert to $\langle S_k \rangle_G$. The program will be designed to maximize such profits; but in a competitive environment, the market making profit will be very close to zero. Hypothesis **(H.5)** represents the competitive limit where the expected profit is exactly equal to zero. This is in agreement with random walk models of asset prices, which have their basis in the assumption that investors act without bias⁹

This argument for average-price efficiency generalizes the conventional transaction-level model (Hasbrouck, 1991). In the transaction-level model, the offer price is the expected future price conditioned on the arrival of a buy transaction; and the bid price is the expected future price conditioned on the arrival of a sell. Our argument replaces the single transaction of Hasbrouck's model with an aggregate order flow imbalance over an interval, and replaces the

⁹ As discussed in Hypothesis **(H.2)**, this assumption could be controversial for other schools of thought.

task of pricing a quote with the task of adjusting the configuration parameters of a market-making program.

Note that a hidden order that changes participation rate unpredictably will alter future price outcomes in a way that cannot be anticipated by market makers. Therefore, there is no reason to expect that an efficiency condition should hold for non-uniform execution schedules.

1.2 The Impact Model

We next derive an impact model from the hypotheses above and its simplified form to first order at each step k .

By Hypothesis (H.4), market impact at the interval k of a variable participation rate trajectory, $\langle \tilde{S}_k - S_{k-1} \rangle_G$, depends only on the current participation rate, π_k , and total number of shares filled, ξ_k . Therefore, it might be modeled as the market impact of a *constant* participation trajectory, $\langle \tilde{S}_j - S_{j-1} \rangle_G^C$, executed with rate π_k , which has filled the same number of shares to a step j not necessarily equal to k . Then, we write:

$$\langle \tilde{S}_k - S_{k-1} \rangle_G = \langle \tilde{S}_j - S_{j-1} \rangle_G^C, \quad (1.1)$$

$$\text{Such that:} \quad \xi_k = \xi_j^C. \quad (1.2)$$

Here, ξ_j^C is the size executed by the constant participation rate trajectory and, from the formula $\xi_j \stackrel{\text{def}}{=} \sum_{i=1}^j \frac{1}{\pi_i}$, it is $\xi_j^C = j \pi_k^{-1}$.

Figure 4 shows this concept visually and includes an example.

Hypotheses (H.3) and (H.5) can be solved recursively for the impact of a *constant* participation trajectory, leading to

$$\langle \tilde{S}_j - S_{j-1} \rangle_G^C = \frac{\varepsilon (1 - p_1)}{(j - 1) \sum_{i=j}^{M \rightarrow \infty} p_i} \langle \tilde{S}_1 - S_0 \rangle_G^C, \quad j \geq 3. \quad (2.1)$$

Here $p_i \stackrel{\text{def}}{=} p(i = [N^*])$ is the probability that the hidden order stops at the end of the interval i , and ε is a constant with $\varepsilon \neq \frac{p_1}{1-p_1}, 0$.

Details on the derivation of equation (2.1) and a discussion of the case where $j = 2$ can be found in Appendix A.

For a Pareto distribution of hidden order sizes, $p_i = \frac{p_1}{i^{\alpha+1}}$, the sum in the denominator, $\sum_{i=j}^{M \rightarrow \infty} p_i$, is proportional to a Hurwitz zeta function $\zeta(\alpha + 1, j)$ as:

$$\sum_{i=j}^{M \rightarrow \infty} p_i = p_1 \zeta(\alpha + 1, j), \quad (2.2)$$

Using the series of equalities (1), (2) and Hypothesis (H.4) for the first interval impact, we write:

Figure 4. Interval–impact local properties. Upper table shows a trajectory schedule with varying participations rates $\{\pi_i\}_{i=1}^k$, which fills ξ_k total transactions in a transactional time t_k . Lower table shows a constant participation rate schedule in j steps that also fills a total of ξ_k institutional transactions with constant rate π_k . Hypothesis 4 asserts that both trajectories have the same interval–impact at k and j respectively because both have the same participation rate at the last interval and fill the same amount of total institutional transactions. This is $\langle \bar{S}_k - S_{k-1} \rangle = \langle \bar{S}_j - S_{j-1} \rangle^C$, where C denotes the constant participation rate trajectory.

The example shows a varying trajectory with $\pi_{k=7} = 0.1$ that fills $\xi_7 = 40$ transactions. A constant trajectory with participation rate $\pi^C = 0.1$ will fill $\xi_j^C = 40$ transactions in $j = 4$ steps, in a longer transactional time, and will have the same interval– impact at the 4th interval than the one of the varying trajectory at the 7th.

Interval i	1	2	3	...			k
τ_i	π_1^{-2}	π_2^{-2}	π_3^{-2}				π_k^{-2}
Transactional time t_i							$t_k = \pi_k^{-2} + \sum_{i=1}^{k-1} \pi_i^{-2}$
Institutional transactions ξ_i							$\xi_k = \pi_k^{-1} + \sum_{i=1}^{k-1} \pi_i^{-1}$
Example	π_i^{-1}	7	5	3	5	6	4
$k = 7$	t_i	49	74	83	248		348
	ξ_i	7	12	15	30		40

Interval i	1	2	...		j
τ_i^C	π_k^{-2}	π_k^{-2}			π_k^{-2}
Transactional time					$t_j^C = j \pi_k^{-2}$
Institutional transactions					$\xi_k = j \pi_k^{-1}$
Example	π_i^{-1}	10	10	10	10
$j = 4$	t_i	100	200	300	400
	ξ_i	10	20	30	40

$$\langle \tilde{S}_k - S_{k-1} \rangle_G = \frac{(\varepsilon/p_1)(1-p_1)}{(j-1)\zeta(\alpha+1, j)} (\hat{\mu}(\sigma) \pi_k^\gamma), \quad (3.1)$$

Here, j is found by inverting condition $\xi_j^C = j \pi_k^{-1}$ from (1.2), as¹⁰:

$$j = \xi_k \pi_k, \text{ and } j \geq 3. \quad (3.2)$$

Given values for ξ_k and π_k , if $j = \xi_k \pi_k = 2$ then $k \neq 1$, and the formula for impact can be found in the Appendix A. The case $j = \xi_k \pi_k = 1$ reduces to $k = 1$, which was defined in Hypothesis **H.4.1** for the first interval impact. From the condition (3.2), we obtain that formula (3.1) is valid when $\xi_k \geq 3 \pi_k^{-1}$, or equivalently, $\xi_{k-1} \geq 2 \pi_k^{-1}$, and $k \geq 3$ in order (3.1) to be also valid for constant participation schedule, where $\xi_{k-1} = (k-1)\pi^{-1}$. In Section 2.2, we will obtain execution schedules that fulfill this condition for all $k \geq 3$, and for $k = 2$, $\text{Minimum}\{\pi_i, 1 \leq i \leq N^*\} > \text{error} = 2\pi_1 - \pi_2 > 0$.

The equation (3.1) is the exact form for the interval-impact when $\xi_{k-1} \geq 2 \pi_k^{-1}$ is fulfilled for $k \geq 3$. We can exactly write the solution to our model, with parameters defined below, as:

$$\begin{aligned} \text{Impact}_k(\{\pi_j\}_{j=1}^k) &= \langle \tilde{S}_k - S_{k-1} \rangle_G = \\ &= \frac{\mu(\sigma) \alpha^{-1}}{(\xi_k \pi_k - 1) \zeta(\alpha + 1, \xi_k \pi_k)} \pi_k^{\beta - \alpha + 1}, k \geq 3. \end{aligned} \quad (4.1)$$

Here, we set $\beta \stackrel{\text{def}}{=} \gamma + \alpha - 1$, and $\mu(\sigma) \stackrel{\text{def}}{=} \left(\frac{\varepsilon}{p_1}\right) (1 - p_1) \alpha \hat{\mu}(\sigma)$. Given $\gamma = -0.2$, from **H.4.1**, and $\alpha = 1.5$, (see Appendix A), we have $\beta = 0.3$.

¹⁰ This is also valid when the product $\xi_k \pi_k$ is not an integer.

From the asymptotic form of the Hurwitz zeta function for large j , $(j-1)\zeta(\alpha+1, j) \xrightarrow{j \gg 1} \alpha^{-1}j^{-\alpha+1}$. **Figure 5.1** shows this behavior.

Therefore, we approximately write:

$$Impact_k(\{\pi_j\}_{j=1}^k) = \langle \tilde{S}_k - S_{k-1} \rangle_G \approx \mu(\sigma)\pi_k^\beta \xi_k^{\alpha-1}, \quad k \gg 1. \quad (4.2)$$

The market impact in equation (4.1) only involves the most recent participation rate and shares acquired through the end of interval k and is valid for all $k \geq 3$. For simplicity, even though equation (4.2) is an approximation for large k , we will use it for all the intervals $k \geq 1$. The difference $(j-1)\zeta(\alpha+1, j) - \alpha^{-1}j^{-\alpha+1}$, is shown in **Figure 5.2** for all $j \geq 3$. For $j > 5$ and $\alpha = 1.5$, it is less than 0.02. Additionally, using the optimal trajectories obtained in Section 2.2, we will confirm that the differences ΔI_k between the impacts of formulas (4.1) and (4.2) verify that

$$\Delta I_{k+1} < \Delta I_k < \Delta I_3 \sim 10^{-1} \times Impact_3, \forall k \geq 4,$$

where $Impact_3$ is calculated by using the exact formula (4.1).

The formula (4.2) is in agreement with the empirical data for constant-participation trade segments, as we showed in the regression (h.4.2) presented in the justification of **H.4.1**. The parameters $\beta = 0.3$ and $\alpha = 1.5$, are consistent with the speed exponent $b_2 = 0.3$ and trade size exponent $b_4 = 0.5$.

Figure 5.1. The functions $(j-1)\zeta(\alpha+1, j)$, represented by the dotted line, and $\alpha^{-1}j^{-\alpha+1}$, where $\alpha = 1.5$ and $\zeta(\alpha+1, j)$ is the Hurwitz Zeta function. We see the overlap for $j \geq 3$. The difference between these two functions is shown in Figure 5.2.

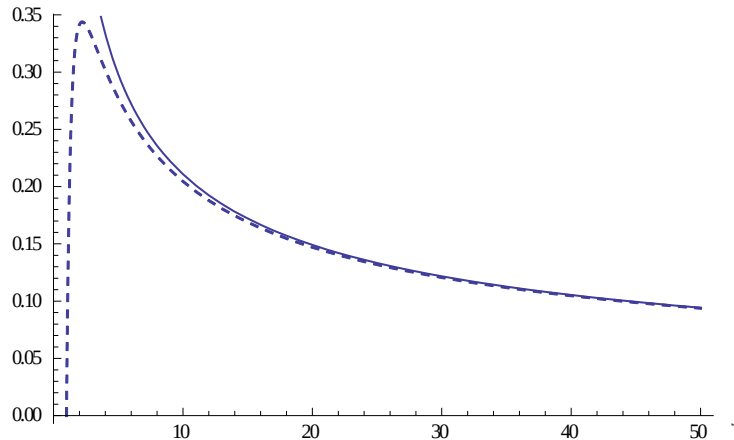
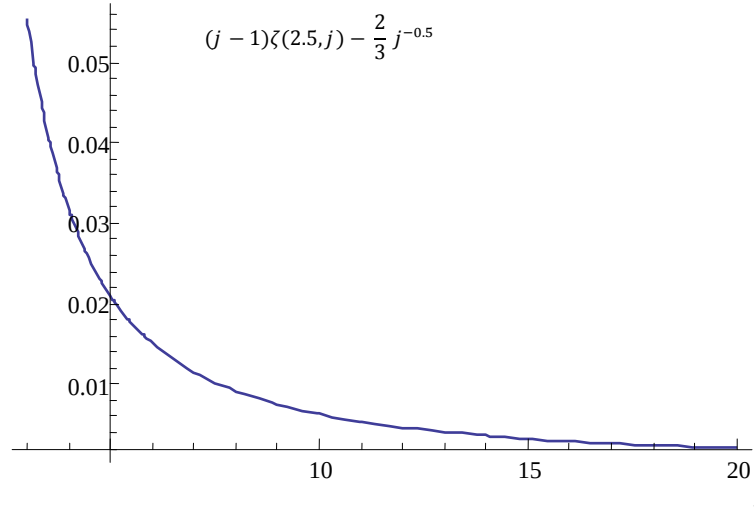


Figure 5.2. The difference $(j - 1)\zeta(\alpha + 1, j) - \alpha^{-1}j^{-\alpha+1}$, for all $j \geq 3$, and $\alpha = 1.5$, where $\zeta(\alpha + 1, j)$ is the Hurwitz Zeta function.



Comments:

- We mentioned before that impact of equation (4.2) may be written as the one of a constant participation rate trajectory π_k that fills ξ_k as $j \pi_k^{-1}$, then:

$$\langle \tilde{S}_k - S_{k-1} \rangle_G \approx \mu(\sigma) \pi_k^{\beta-\alpha+1} j^{\alpha-1}. \quad (4.3)$$

The above formula depends on the *local* participation rate π_k and does not depend on *how* the shares were filled; then, *interval–impact does not depend of the history of the trade*.

Note that, given ξ_k and π_k , any schedule $\{\pi_i\}_{i=1}^{k-1}$ such that $\xi_k - \pi_k^{-1} = (\sum_{i=1}^{k-1} \pi_i^{-1})$ will have the same interval–impact; this includes any of the $(k-1)!$ trajectories that correspond to the permutations of the set $\{\pi_i\}_{i=1}^{k-1}$.

- Equation (4.2) shows that *interval–impact depends of the length of the interval to the origin of the trade*. This can be easily seen if the trade is realized with a constant participation rate π . In such a case, equation (4.2) becomes

$$\langle \tilde{S}_k - S_{k-1} \rangle_G \approx \mu(\sigma) \pi^{\beta-\alpha+1} k^{\alpha-1}. \quad (4.4)$$

Therefore, *interval–impact is non-stationary*.¹¹

1.3 Price Dynamics

Following the ideas given above, explicitly equations (h.2.4) and formula (4.2), we will write the evolution of the security price as:

¹¹ In the sense that Impact is explicitly time-dependent. In equation (4.4), the step k may be written as a function of time.

$$\tilde{S}_k = S_{k-1} + \mu \pi_k^\beta \xi_k^{\alpha-1} + \sigma \tau_k^{\frac{1}{2}} \varsigma_k, \quad [N^*] \geq k \geq 1. \quad (5.1)$$

Here, $[\mu] = \frac{\$}{\text{share}}$, $\mu > 0$ for a buy and $\mu < 0$ for a sell. As explained in **H.2.2**, the $\{\varsigma_k\}$ are random variables with zero Gaussian mean and unit variance, and σ is a constant with units $[\sigma] = \frac{\$}{\text{share} \times \sqrt{\text{transaction}}}$.

By Hypothesis (**H.1**), we are considering the possibility that the total number of steps N^* be a non-integer value; which means the institution may require a fraction q , where $q = N^* - [N^*]$, $0 \leq q < 1$, of what would otherwise become a detectable interval to complete the trade. In the case where $q \neq 0$, the price at N^* will be¹²:

$$\tilde{S}_{N^*} = S_{[N^*]} + \mu \pi_f^\beta \xi_{N^*}^{\alpha-1}, \quad (5.2)$$

where ξ_{N^*} is the total number of transactions traded until the last interval N^* , and it is by definition:

$$\xi_{N^*} \stackrel{\text{def}}{=} \left(\sum_{i=1}^{[N^*]} \frac{1}{\pi_i} + q \pi_f^{-1} \right). \quad (6)$$

Combining (5) with the breakeven Hypothesis (**H.3**), we derive by recursion the expression for the execution price at k , as a function of the participation rate schedule π_i , $1 \leq i \leq k$:

$$\tilde{S}_k = S_0 + \mu \left\{ \pi_k^\beta \xi_k^{\alpha-1} + \sum_{i=1}^{k-1} \pi_i^{\beta-1} \xi_i^{\alpha-2} \right\} + \sigma \sum_{i=1}^k \pi_i^{-1} \varsigma_i, \quad 2 \leq k \leq [N^*], \quad (7.1)$$

$$\tilde{S}_1 = S_0 + \mu \pi_1^{\beta-\alpha+1} + \sigma \pi_1^{-1} \varsigma_1, \quad (7.2)$$

$$\tilde{S}_{N^*} = S_0 + \mu \left\{ \pi_f^\beta \xi_{N^*}^{\alpha-1} + \sum_{i=1}^{[N^*]} \pi_i^{\beta-1} \xi_i^{\alpha-2} \right\} + \sigma \sum_{i=1}^{[N^*]} \pi_i^{-1} \varsigma_i. \quad (7.3)$$

The reader may refer to the Appendix B for the proof of the above derivations. In these formulas, total impact from steps 0 to k is the term containing $\mu \{ \}$. Two summands contribute to the total impact. In one term, we see the explicit dependence with the last participation rate π_k and the total shares ξ_k transacted to step k while in the other one, we find the sum of partial contributions to one step before k and each of them in agreement with Hypothesis (**H.4**). This concept is written as:

¹² Note that we depreciate noise in that fractional step because of the shorter contribution in time in a low volatility environment

$$Total\ Impact(k, 0) \stackrel{\text{def}}{=} \langle \tilde{S}_k - S_0 \rangle_G = \mu \left\{ \pi_k^\beta \xi_k^{\alpha-1} + \sum_{i=1}^{k-1} \pi_i^{\beta-1} \xi_i^{\alpha-2} \right\},$$

alternatively, the same:

$$Total\ Impact(k, 0) \stackrel{\text{def}}{=} \langle \tilde{S}_k - S_0 \rangle_G = \langle \tilde{S}_k - S_{k-1} \rangle_G + \sum_{i=1}^{k-1} \frac{\langle \tilde{S}_i - S_{i-1} \rangle_G}{\pi_i \xi_i}.$$

Comments:

- The formula above indicates that *total impact* of the trade to step k depends on the *interval-impact* of every other step before, then it depends of the history of the trade until k as we see the explicit dependency on the participation rate schedule $\{\pi_1, \pi_2, \dots, \pi_k\}$. For example, two trajectories $\{\pi_1, \pi_2, \pi_3\}$ and $\{\pi_2, \pi_1, \pi_3\}$ fill the same amount of shares in three steps but have different total impacts.
- Furthermore, total impact is non-stationary. This result is readily seen for a constant trajectory π , where $\xi_i = i \pi^{-1}$. We get:

$$\langle \tilde{S}_k - S_0 \rangle_G = \mu \pi^{\beta-\alpha+1} \left\{ k^{\alpha-1} + \sum_{i=1}^{k-1} i^{\alpha-2} \right\},$$

which means that

$$Total\ Impact(k, 0) = f(k, k-1).$$

The explicit dependency on the steps makes impact non-stationary.

- It is straightforward to arrive to the expressions:

$$Total\ Impact(k, 0) = Total\ Impact(k-1, 0) + \mu \left\{ \pi_k^\beta \xi_k^{\alpha-1} + \pi_{k-1}^{\beta-1} \xi_{k-1}^{\alpha-2} (1 - \pi_{k-1} \xi_{k-1}) \right\},$$

$$\langle \tilde{S}_k \rangle_G = \langle \tilde{S}_{k-1} \rangle_G + \mu \left\{ \pi_k^\beta \xi_k^{\alpha-1} + \pi_{k-1}^{\beta-1} \xi_{k-1}^{\alpha-2} (1 - \pi_{k-1} \xi_{k-1}) \right\}.$$

Note that the dynamic of prices is Markovian because it can be calculated at step k in terms of the immediate previous step and is not influenced by the information of $\langle \tilde{S}_i \rangle_G$ for $i < k-1$.

2. Optimal Execution

Our objective is to obtain an optimal execution schedule $\{\pi_1, \pi_2, \dots, \pi_{[N^*]}, \pi_f, N^*\}^{optimum}$ that minimizes the *risk-adjusted cost function*, which is defined below.

2.1 The Risk and Cost of the Trade

We call the *capture of a trajectory* the complete trading revenue; namely, this is

$$Capture \equiv \sum_{i=1}^{[N^*]} n_i \tilde{S}_i + q n \pi_f^{-1} \tilde{S}_{N^*},$$

where $n_i = n \pi_i^{-1}$ is the number of shares traded in the i – interval and n is the number of traded shares in each institutional transaction with $n > 0$ for a sell and $n < 0$ for a buy.

The *total cost of the trade*, $E(\vec{\pi}, N^*)$, is the difference between the initial value of the total fill and the capture of the trajectory. The corresponding equation is:

$$E(\vec{\pi}, N^*) \stackrel{\text{def}}{=} n \xi_{N^*} S_0 - Capture. \quad (8)$$

Additionally, we will evaluate the *variance of the cost* $V(\vec{\pi}, N^*) \stackrel{\text{def}}{=} \langle (E(\vec{\pi}, N^*) - \langle E(\vec{\pi}, N^*) \rangle_G)^2 \rangle_G$. This function is important to consider because of its relation with the *risk of a trade*.

After a calculation that can be found in the Appendix C, using equations (7), the *expected total cost* turns out to be:

$$\langle E(\vec{\pi}, N^*) \rangle_G = |\mu n| \xi_{N^*} \left\{ \sum_{k=1}^{[N^*]} \pi_k^{\beta-1} \xi_k^{\alpha-2} + q \pi_f^{\beta-1} \xi_{N^*}^{\alpha-2} \right\}, \quad (9)$$

while the *variance* of $E(\vec{\pi}, N^*)$ takes the form

$$V(\vec{\pi}, N^*) = \sigma^2 n^2 \sum_{k=1}^{[N^*]} \pi_k^{-2} (\xi_{N^*} - \xi_{k-1})^2. \quad (10)$$

Note that $\langle E(\vec{\pi}, N^*) \rangle_G > 0$ always, either a buy or a sell.

Having defined the total cost of the trade and its variance, we are able to proceed as in (Almgren and Chriss, 2000), and minimize our objective function that will be a linear combination of the total cost and its variance or, the same, the *risk-adjusted cost function*. The purpose of the optimization of the sum of both quantities is to obtain a trajectory that strives to minimize both the cost and the execution risk.

We next write the *risk-adjusted cost function*:

$$U(\vec{\pi}, N^*; \vec{p}) \stackrel{\text{def}}{=} \langle E(\vec{\pi}, N^*) \rangle_G + \lambda V(\vec{\pi}, N^*). \quad (11)$$

Here, λ is the *risk parameter* with units $[\lambda] = \$^{-1}$, and $\vec{p} = (\lambda, \mu, n, \sigma, \alpha, \beta)$ is a parameter vector defining the parameters of the trade. We remind that we will be using $\alpha = 1.5$, $\beta = 0.3$, as explained in Section 1.2.

For simplicity, we suppose that there exists $N \in \mathbb{N}, N \leq N^*$, such that from $N + 1$ to N^* the institution participates with a constant rate π_f . Therefore, the variables $(\vec{\pi}, N^*)$ will be reduced to $(\{\pi_i\}_{i=1}^N, \pi_f, N^*)$.

We define the *dimensionless risk-adjusted cost* $\frac{U(\vec{\pi}, N^*; N, L)}{|\mu n|}$, and the *dimensionless risk-constant* $L = \lambda \sigma^2 |n/\mu|$, which allows an independent solution on the selection of the parameters n and μ .

Applying the previous expressions, we obtain our objective function, the *dimensionless risk-adjusted cost* with variables $(\vec{\pi}, N^*) = (\{\pi_i\}_{i=1}^N, \pi_f, N^*)$, to be optimized:

$$\begin{aligned} \frac{U(\vec{\pi}, N^*; N, L)}{|\mu n|} = & \xi_{N^*} \left\{ \sum_{k=1}^{[N^*]} \pi_k^{-0.7} \left(\sum_{i=1}^k \pi_i^{-1} \right)^{-0.5} + (N^* \right. \\ & \left. - [N^*]) \pi_f^{-0.7} \left(\sum_{i=1}^{[N^*]} \pi_i^{-1} + (N^* - [N^*]) \pi_f^{-1} \right)^{-0.5} \right\} \\ & + L \sum_{k=1}^{[N^*]} \pi_k^{-2} \left(X - \sum_{j=1}^{k-1} \pi_j^{-1} \right)^2. \end{aligned} \quad (12)$$

Here, $\pi_k = \pi_f, N + 1 \leq k \leq N^*$.

We will set the constraints on the total number of institutional transactions

$$X = \xi_{N^*} \stackrel{\text{def}}{=} \left\{ \sum_{j=1}^N \pi_j^{-1} + (N^* - N) \pi_f^{-1} \right\}, \quad (13)$$

and a maximum for the trading time t_{N^*} :

$$T_M \geq t_{N^*} \stackrel{\text{def}}{=} \sum_{i=1}^N \pi_i^{-2} + (N^* - N) \pi_f^{-2}. \quad (14)$$

2.2 Risk Adjusted Cost Optimization

In what follows, we will focus on finding the optimal trading trajectories for a theory with varying participation rate and various values of risk aversion. By optimal trading trajectories, we mean participation rate schedules that minimize the total risk-adjusted cost function (12), with the constraints (13) and (14). The order of each execution is important because the function (12) is not symmetric under permutations of its arguments $(\{\pi_i\}_{i=1}^N, \pi_f)$.

Results:

The results are summarized in **Table II** for the values of the constraints $X = 100$ lots, $T_M = 1000$ transactions, the parameters $\alpha = 1.5, \beta = 0.3, N = 10$, and for different values of the risk constant L . The optimal trajectories $\vec{\pi}$, are shown in the **Figures 6 to 10**.

L	$\frac{\langle E \rangle}{ \mu n }$	$\frac{\lambda}{ \mu n } V$	$\frac{U}{ \mu n }$	N^*	t_{N^*}
3×10^{-4}	925.65	527.63	1453.29	13.14	1000
1×10^{-4}	827.97	240.12	1068.09	10.99	1000
0	755.74	0	755.742	11.57	1000

Table II. Optimal Risk Adjusted Cost. We show results for the execution cost and risk for $X = 100$ lots, $T_M = 1000$ transactions, $\alpha = 1.5, \beta = 0.3, N = 10$, for the different values of the risk constant $L = \lambda \sigma^2 |n/\mu|$. The second column is the expected total cost, the third gives the total risk, the fourth is the total risk-adjusted cost. N^* is the total number of detectable intervals realized by the hidden order. The last column indicates that the institution reaches the maximum limit T_M of the transactional time. All values are dimensionless; the optimal trajectories $(\vec{\pi}, N^*)$ are independent of the parameters μ and n . An optimal solution is derived from a value of L and the constraints on available time and number of lots.

Below, we graph the participation rate π_k versus the detectable step k , in a continuum approximation, for the different values of the risk - constant $L = \lambda \sigma^2 |n/\mu|$.

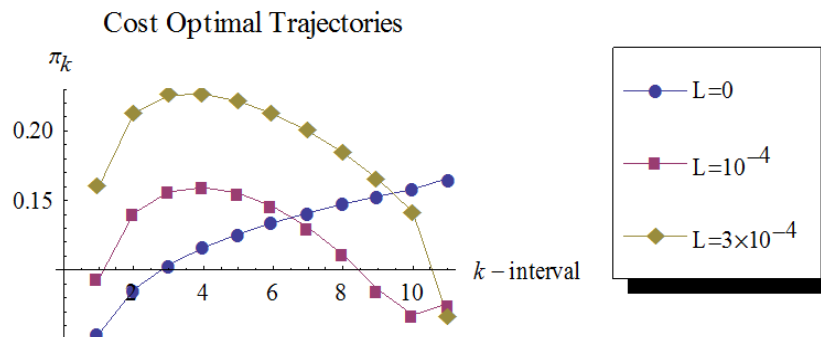


Figure 6. Optimal trajectories representing the participation rate in function of the number of the detectable interval for different values of the risk constant.

On average, the participation rate should be constant in each step. We present a graph of the participation rate versus the transactional time, $t_k = \sum_{i=1}^k \pi_i^{-2}$, corresponding to each k – interval, for each L . The interpretation of each trajectory is in the figure caption.

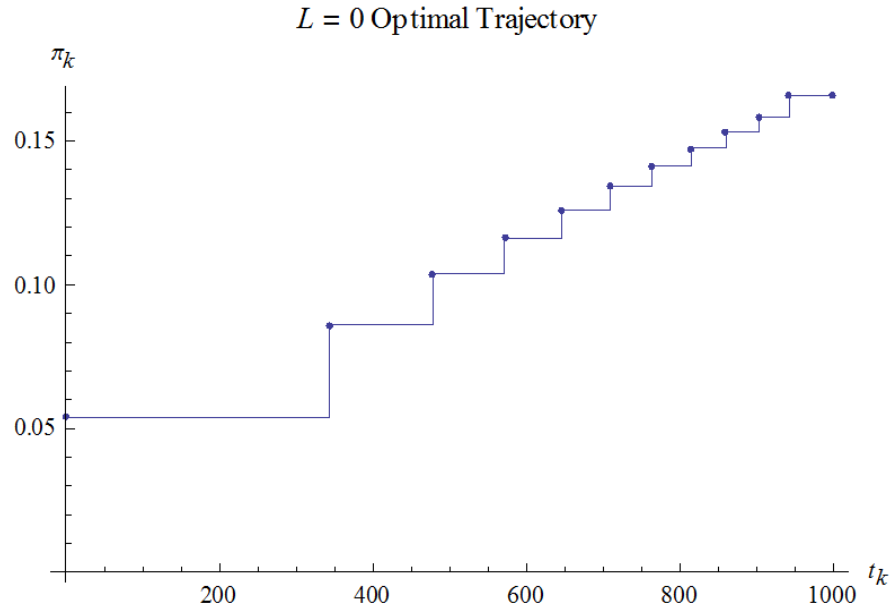


Figure 7. The optimal participation rate schedule is shown as a function of the transactional time, $t_k = \sum_{i=1}^k \pi_i^{-2}$, corresponding to each k – interval, with zero risk aversion. This result shows that, in absence of risk aversion, it is optimal to start the trade more slowly to minimize information leakage, and impact, early in the trade.

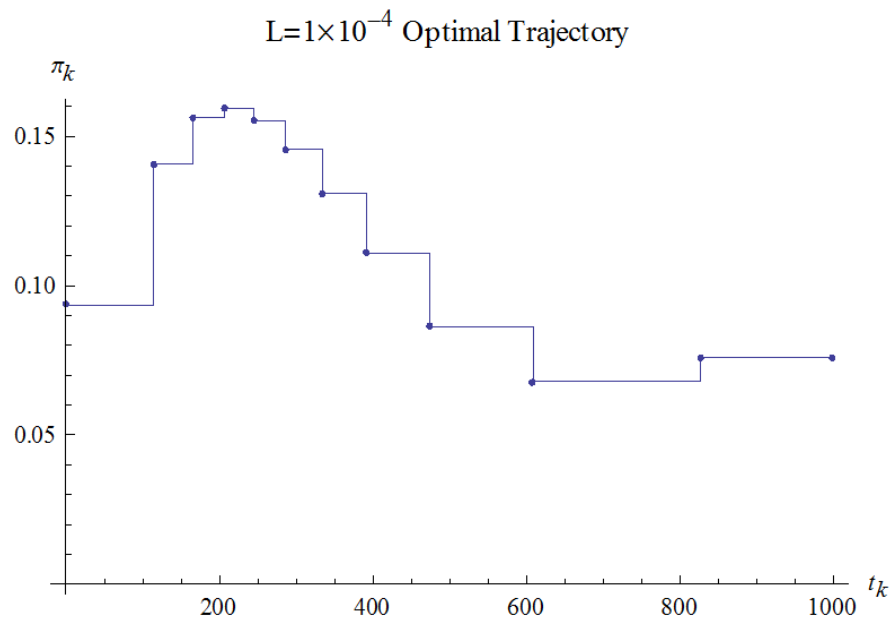


Figure 8. The optimal participation rate schedule is shown as a function of the transactional time, $t_k = \sum_{i=1}^k \pi_i^{-2}$, corresponding to each k – interval, considering risk aversion $L = 1 \times 10^{-4}$. Risk minimization requires trading more in the first half of the allowed time. This is because risk, at each step k , depends of the transactional interval π_k^{-2} and of the square of number of transactions that remains to complete $(X - \sum_{j=1}^k \pi_j^{-1})$. A slow start at the beginning of the trade leaves more transactions to be filled and this will increase the risk. On the other hand, going faster while transacting the same amount of shares would increase impact, as shown in formula (4.2). The figure shows that the objective of impact reduction dominates in the first step. The participation rate is increased in the next eight steps to reduce risk. Near the end, when the institutional transactions or shares are almost exhausted, impact cost dominates over risk and smaller participation rates are optimal.

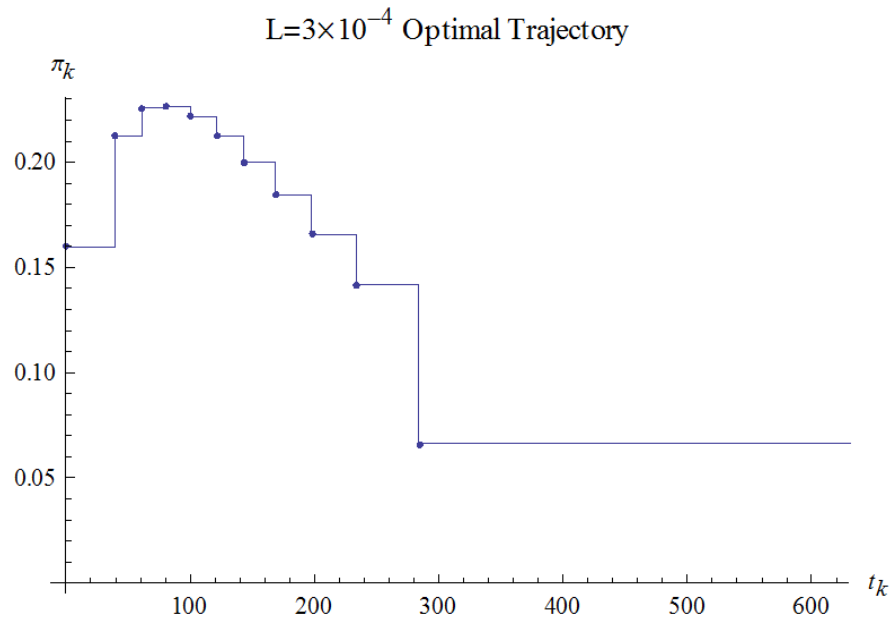


Figure 9. The optimal participation rate schedule for a very risk averse trader is shown as a function of the transactional time, $t_k = \sum_{i=1}^k \pi_i^{-2}$, corresponding to each k – interval, for risk aversion $L = 3 \times 10^{-4}$. In comparison with Figure 8, the front-loading of the execution is more pronounced because the risk aversion is higher, but the soft start remains optimal. The faster and deeper decrease in participation rate towards the end compensates for a bigger impact that arises from implementing higher rates at the beginning of the trade. The long tail at constant rate appears, in part, for the additional constraint we imposed that $\pi_i = \text{constant}$ for $i \geq 11$. This constraint was a restriction on the space of solutions intended to simplify the optimization problem. The high rates at the beginning lead to smaller intervals and number of transacted shares, then more intervals are needed to complete the trade. Without the restriction, one could likely find slightly lower cost solutions with a larger total number of intervals. The resulting trajectory would then most likely continue with a step descent as in Figure 8.

Below, we draw the comparative graph of optimal schedules as a smoothed function of transaction time for the different values of the risk aversion.

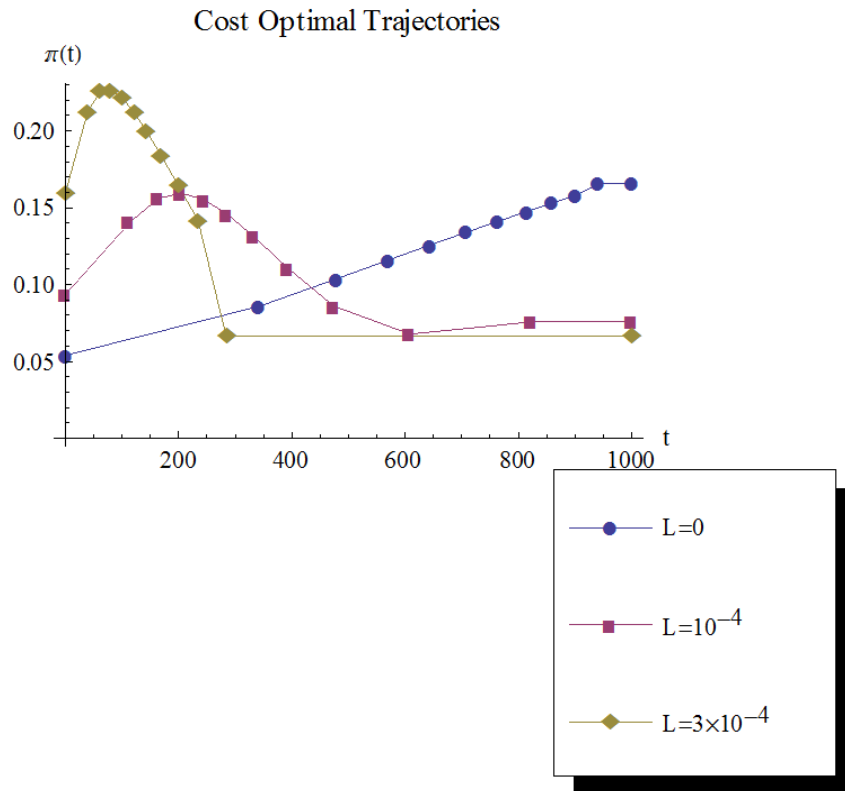


Figure 10. Comparative graph of optimal trajectories as smoothed functions of the transactional time, for the different values of the risk aversion. The dotted line gives the “equilibrium” or optimal solution for the cost that only accounts for impact. Slow increments in participation rates minimize impact. However, when risk aversion is introduced, the optimal schedules require increasing participation rates at the beginning compensated by a descent at the end. In the most risk-averse case (diamonds), the drop in participation rate after front-loading is abrupt. As explained in the previous figures, the optimal schedule reflects a compromise between two objectives. Risk minimization requires trading more in the first half of the allowed time whereas the goal of minimizing market impact requires a limited participation rate, at the very beginning of the trade, to avoid an information shock from early detection.

3. Optimizing Performance to the VWAP Benchmark

In the preceding section, we optimized the sum of the expected implementation shortfall and its variance. We called it the risk-adjusted cost function and we found optimal trading trajectories.

The implementation shortfall is the difference between the initial book value of the shares, XS_0 , and the capture of the trajec-

tory $\sum_{i=1}^{N^*} n_i \tilde{S}_i$. Nevertheless, traders have other objectives or benchmarks rather than XS_0 . These are:

- 1) Post reversion price or closing price. This is useful to measure the effect of an entry trade on assets under management (AUM). If a trade is profitable when marked to market after reversion, its immediate effect on AUM is positive. In this paper, we showed that this measure of profitability is null on average, so there is no optimization problem with regard to the closing price. One can think of realizations with positive mark to market profits as the winners from the perspective of the asset manager's information. The possibility of predicting the information content of a trade, and using this to optimize the execution schedule, is the subject of a separate paper (Criscuolo and Waelbroeck, 2012).
- 2) Volume-weighted average price during the transactional period, or VWAP. This is the average price transacted by the market. This benchmark is often used to evaluate the effectiveness of an algorithmic execution strategy, in cases where there is no urgency. A better-than-VWAP execution reveals a success of the algorithm at selecting more favorable price points.

Both, the closing price and the VWAP, are affected by the same market impact that also affects the execution price, so the difference between the benchmark and the execution price fails to account for impact costs. As a result, a better-than-VWAP execution will not necessarily be the lowest cost strategy. In what follows, we compare the optimal cost trajectories to VWAP-optimal ones to understand to what extent VWAP optimization conflicts with the objective of minimizing implementation shortfall.

We will analyze the difference Δ between VWAP and the capture:

$$\begin{aligned} \Delta(\text{buy or sell}) &:= \mp \text{VWAP} \pm \text{capture} \stackrel{\text{def}}{=} \\ &\stackrel{\text{def}}{=} \mp \frac{1}{t_{N^*}} \sum_{i=1}^{N^*} \tau_i \tilde{S}_i \pm \frac{1}{\xi_{N^*}} \sum_{i=1}^{N^*} \pi_i^{-1} \tilde{S}_i. \end{aligned} \quad (15)$$

Here, t_{N^*} is the institutional trade duration and ξ_{N^*} is the total number of institutional orders executed in that period. $\{\pi_i\}_{i=1}^{N^*}$ is the set of participation rates from the first to the last interval and $\tau_i = \pi_i^{-2}$ is the minimum expected transactional market period for the detection of the i^{th} interval. π_i^{-1} is the number of the institutional transactions executed in the $i - detectable\ interval$, and $\tilde{S}_i$ is the market price per share paid in the $i - interval$.

Following the equations (7), and taking $\mu(\text{sell}) = -\mu(\text{buy})$ equation (15) is:

$$\begin{aligned}
 & \langle \Delta(\text{buy or sell}, \{\pi_i\}_{i=1}^{N^*}) \rangle_G \\
 &= -\frac{|\mu|}{t_{N^*}} \left\{ \sum_{k=1}^{[N^*]} \pi_k^{\beta-1} \xi_k^{\alpha-2} (\pi_k^{-1} \xi_k - t_k) \right. \\
 & \quad \left. + q \pi_f^{\beta-1} \xi_{N^*}^{\alpha-2} (\pi_f^{-1} \xi_{N^*} - t_N^*) \right\}, \tag{16}
 \end{aligned}$$

where $\xi_k \stackrel{\text{def}}{=} \sum_{i=1}^k \pi_i^{-1}$ and $t_k \stackrel{\text{def}}{=} \sum_{j=1}^k \tau_j$.

Next, we minimize (16) using Mathematica8 constrained with $X = \xi_{N^*} = 100$, $t_{N^*} \leq T_M = 1000$. The optimal solution is:

$$\Delta(\text{buy or sell}, \text{minimum})/|\mu| = -0.666758,$$

with $N^* = 10.97$, $t_{N^*} = 1000$ and the optimal trajectory shown in **Figure 11**.

The negative value indicates that the institution buys below the market or sells above the market, and the gain (absolute value) is maximal for the optimal trajectory.

The cost for the solution, which optimizes execution performance relative to the VWAP, is $\frac{E}{|n\mu|} = 837.36$. This is higher than the optimal cost $\frac{E}{|n\mu|}(\text{minimum}) = 755.74$, for $X = 100, t_{N^*} = 1000, L = 0$, found in Section 2.1. Thus, it is possible to beat the VWAP benchmark with that trajectory but at the expense of a higher shortfall.

Conversely, the value for the VWAP benchmark using the optimal trajectory, which minimizes the shortfall, is: $\frac{\Delta(\text{buy or sell}, \text{optimal for } L=0)}{|\mu|} = 3.21403$. The positive value means that the institution reduces the shortfall to the minimum at expenses of selling below or buying above the market average price.

Below, we show the graph of the participation rate versus transactional time for the VWAP-optimal solution:

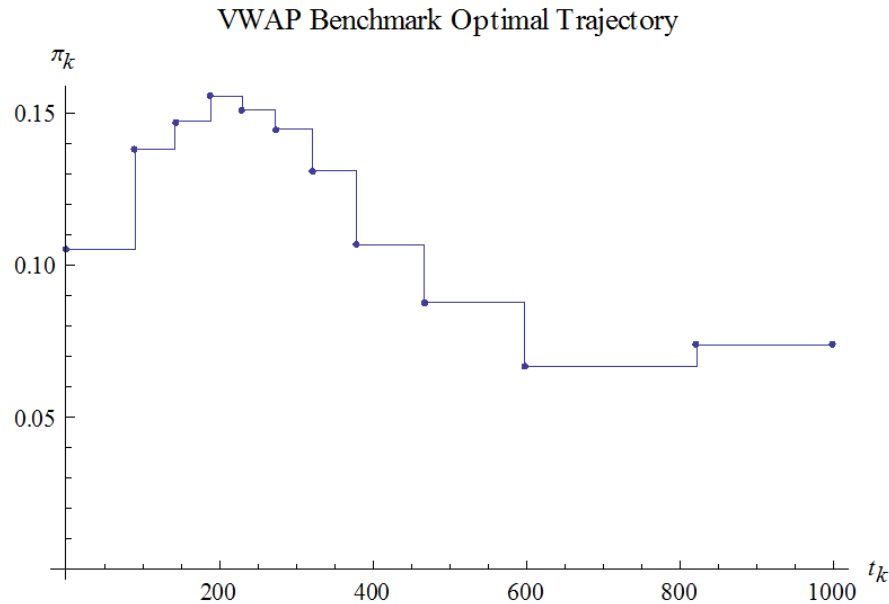


Figure 11. The participation rate schedule, which maximizes the average price performance relative to the VWAP benchmark, is shown as a function of the transactional time, $t_k = \sum_{i=1}^k \pi_i^{-2}$, corresponding to each k – interval. This optimal solution is similar to the optimal risk averse schedule, because the optimal strategy to beat a VWAP benchmark is to capture more of the transactions at the beginning of the trade, which is also what minimizes risk. However, front-loading the execution schedule increases expected cost. In this example, the additional cost that must be paid for VWAP optimization is approximately 10% of the optimal cost.

4. Discussion

We developed a theory to explain market impact as a process wherein market makers observe a hidden order. They estimate it given the distribution of hidden order sizes and the expectation that institutional trades break even after reversion. As explained by Hasbrouck (Hasbrouck, J. 1991), only the unexpected component of order flow can cause incremental market impact. As we showed here, the increasing predictability of the order flow causes the impact function to be concave. *The expected order flow imbalance is incorporated in the security price in the form of temporary impact.* When the trade ends, expectations of further order flow subside and price reverts to a level that incorporates only permanent impact. In contrast, in Kyle's classic paper (Kyle, A. 1985) the informed trader knows the final price and therefore will only buy below this price, so there is no reversion. These models are different because they represent different situations. Portfolio managers often execute orders with no private information for liquidity reasons, or to realize long-term portfolio objectives. There is ample evidence that the algorithmic execution of hidden orders leads to a predictable order flow and a concave impact function (Bouchaud et al, 2008).

We provided solutions that minimize the risk-adjusted cost for various levels of risk aversion or the performance relative to a VWAP benchmark. The optimal risk-averse execution schedules differ from the classic Almgren-Chriss solutions, mainly, in that they do not start the trade at the peak participation rate (see Figure 12). Our theory predicts that the aggressive trade start in the A-C solution will result in greater information leakage and, therefore, higher costs for the rest of the trade. Our model prefers a softer start, immediately followed by a peak, and ending with a constant rate. Below, we will quantify with an example the additional cost that must be paid for risk aversion, and from the information shock at the beginning of the A-C solution.

The VWAP optimal solution is similar to the optimal risk-averse schedule. This should not be surprising – both VWAP optimization and risk reduction require trading more at the beginning. The particular “ocean-wave” shape of the optimal schedule results from the desire to also limit market impact. However, a compromise is necessary. To beat the VWAP, this solution accepts higher impact early in the trade, represented by the peak of the wave, resulting in a higher/lower price (buy/sell) throughout the trade and a higher implementation shortfall.

4.1 Example and Comparisons

We illustrate the differences between the solutions presented in this paper using a specific example. We consider a mid-cap trade of $|X_n| = 25000$ shares, $|n| = 250$ share/transaction, $S_0 = \$50$, with an average participation rate of 10% ($\pi = 0.1$), transactional volatility $\sigma = 0.019$ \$/share and impact parameter $|\mu| = 0.0315$ \$/share.

The shortfall and the VWAP benchmark of risk-adjusted cost optimal solutions are listed in **Table III** for this example and different risk aversion parameters.

$L \times 10^4$	Risk parameter λ ($\$^{-1}$)	Shortfall $\frac{\$}{(share)}$	Shortfall E (\$)	Risk $\sqrt{V}$ (\$)	$\Delta_{VWAP} \frac{\$}{(share)}$
0	0	0.2381	5953.5	9192.27	0.1012
1	3.49×10^{-5}	0.2608	6520.5	7360.83	-0.0173
3	1.05×10^{-4}	0.2917	7292.3	6290.65	0.2418

Table III. Shortfall and VWAP benchmark of Cost Optimal solutions. We consider a mid-cap trade of 25000 shares, in a $S_0 = \$50$ security, executed at an average participation rate of 10% ($\pi = 0.1$). If the security's trading volume is $\frac{400 \text{ transactions}}{hr}$, a detectable interval will represent 15 minutes of trading. The impact for a 15-minute interval is estimated to be 10bps for

this security, i.e. $|\mu| = 0.0315 \text{ \$/share}$. We take an annual volatility of 30% or $\sigma = \frac{0.95(\frac{\$}{\text{share}})}{\sqrt{\text{day}}}$. One day consists of 6 hours and 30 minutes or 2600 *market transactions* in this example. The last column is the VWAP benchmark calculated with the risk adjusted cost optimal solution. The case with moderate risk aversion, $L = 10^{-4}$, captures a price that is \$0.0173 better than VWAP but leads to a total cost that is 9.5% higher than the optimal solution with no risk aversion.

The VWAP-optimal solution is shown in **Table IV**. It is most similar to the risk-averse solution with $L = 1 \times 10^{-4}$.

<i>Model</i>	Δ_{VWAP} \$(\frac{\\$}{\text{share}})\$	<i>Shortfall</i> <i>per share</i> \$(\frac{\\$}{\text{share}})\$	<i>Risk</i> $\sqrt{V}$ \$(\\$/\text{share})\$
<i>VWAP</i> – <i>optimal</i>	–0.0210	0.2638	0.2893 7233.84\$(total)

Table IV. Comparison of optimal VWAP and shortfall. VWAP optimal value is given in the first column, and the shortfall and its variance per share corresponding to the VWAP optimal trajectory in the second and third column respectively.

The Almgren-Chriss solution (A-C) most similar to our VWAP-optimal solution is shown in **Figure 12**. Its cost per share is \$0.2752, representing a higher implementation shortfall by \$0.0114 or 2.3bps.

Cost Optimal Trajectories Comparison

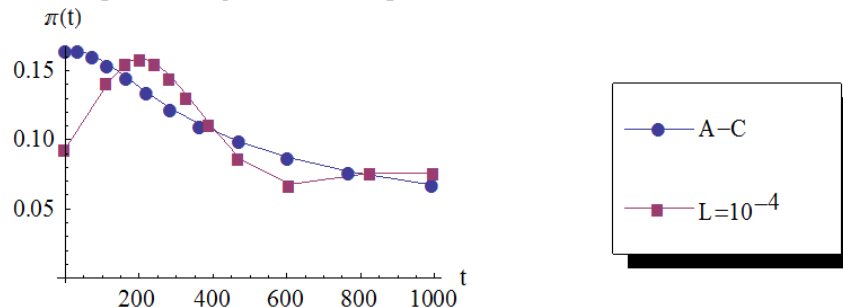


Figure 12. Comparative graph of the cost optimal trajectories as a function of the transactional time. The A-C line is the solution predicted by Almgren-Chriss formulation with linear impact and risk averse constant $\lambda(\text{Almgren} - \text{Chriss}) = 10^{-5}/\text{\$}$. The square-line is the optimal trajectory for the non-linear theory for risk averse constant $L = 10^{-4}$ or $\lambda = 3.49 \times 10^{-5}/\text{\$}$, as shown in Figure 8. The implementation shortfall of the A-C solution at \$0.2752 is 15.6% greater than our optimal solution with no risk aversion, and 5.5% greater than the optimal solution with $L = 10^{-4}$, because of the information shock in the abrupt trade start of the A-C solution.

4.2 Comments about the Hypotheses

Among the assumptions of our theory, the one that hidden orders are detectable stands as, perhaps, the most controversial. It is generally impossible to detect with certainty (or $p = 1$) the existence of a hidden order. However, as we will show next, such an assertion will not modify our results. If one has a detection probability $p < 1$, the efficiency condition becomes an average over three possibilities. Either there is no hidden order and the last interval price equals the expected future price, or there is one, and we have the two possibilities where the hidden order either continues or stops. The efficiency condition becomes:

$$\langle \tilde{S}_k \rangle_G = (1 - p) \langle \tilde{S}_k \rangle_G + p(P(k + 1|k) \langle \tilde{S}_{k+1} \rangle_G + (1 - P(k + 1|k)) \langle S_k \rangle_G) \Leftrightarrow$$

$$P(k + 1|k)(\langle \tilde{S}_{k+1} \rangle_G - \langle \tilde{S}_k \rangle_G) + (1 - P(k + 1|k))(\langle S_k \rangle_G - \langle \tilde{S}_k \rangle_G) = 0,$$

Which is the same as in Hypothesis (H. 5). Therefore, the expression for market impact in a model with finite detection probability remains the same, as do the optimal execution solutions.

In relation to the detection of hidden orders, we used a discretization of the time line into “detectable” intervals comprising $1/\pi^2$ transactions. This choice was based on the argument that the imbalance in order flow should violate the hypothesis of a null bias with $t\text{-stat}=1$, which only provides a minimal level of confidence in detecting a hidden order. In practice, one would expect competing market makers to devise their own detection model. They will be looking not only at order flow, but also at sector-relative returns, signs of repeated order placements, available liquidity in dark pools, etc. The advantage goes to the one who is first able to detect the hidden order, which is why we chose the low value $t\text{-stat}=1$. However, while differences in the hidden order detection mechanism will lead to different discretizations of the execution process, the structure of the model remains the same as long as the detection interval remains proportional to $1/\pi^2$. The shape of the optimal execution solutions described in this paper remains qualitatively similar if one chooses to define a detectable interval with a higher level of confidence. In **Figure 13**, we show the optimal participation rate profile for the non-risk-averse case and $t\text{-stat}=2$. While not identical to Figure 7, the main qualitative feature of a back loaded execution schedule remains intact.

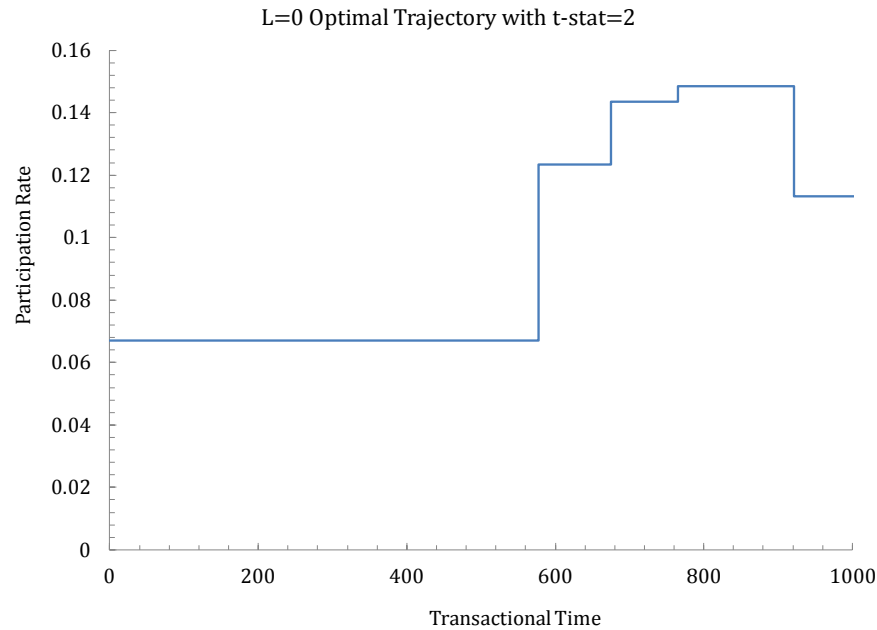


Figure 13. Optimal participation rate profile without risk aversion and with a detection model based on a level of confidence $t\text{-stat}=2$. The optimal solution executes the first 600 transactions with a participation rate of approximately 7%, similar to the solution with $t\text{-stat}=1$ found in Figure 7. The execution back-loading and peak participation rate of 15% is also similar to the previous case. Although, the solutions are different in detail, they are qualitatively similar. The optimal execution framework is robust to minor variations in the assumptions regarding hidden order detection.

Acknowledgements

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The phenomenological results regarding the first-interval impact model and breakeven conditions are part of ongoing research in collaboration with Carla Gomes, which will be published separately.

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Appendix A. Derivation of Market Impact for Constant Participation Rate

Here, we will derive the interval–market impact (2.1) for a constant participation rate schedule. This expression was first obtained in (Farmer et al, 2011) but we will reproduce the derivation for completeness following our particular notations and definitions.

Market participants, who are able to detect the presence of a hidden order at a particular i –interval, will formulate expectations about the remaining size after i . To quantify those expectations, we will define the probability that the hidden order stops at i (or $[N^*] = i$) to be $p([N^*] = i) \equiv p_i$. We also define the probability $p(i + 1|i)$ that the hidden order continues at $(i + 1)$ given that it was detected at the i –interval. Then, by the definition of conditional probability, we write:

$$p(i + 1|i) \stackrel{\text{def}}{=} \frac{p([N^*] \geq i + 1)}{p([N^*] \geq i)} = \frac{\sum_{k=1}^{M \rightarrow \infty} p_{i+k}}{\sum_{k=0}^{M \rightarrow \infty} p_{i+k}}. \quad (\text{A. 1})$$

The second equality comes from the definition of probability of the union of disjoint sets, and M represents the maximum size of any trade in step units. It is natural to take $\lim_{k \rightarrow \infty} p_k = 0$, which means $p_k \approx 0$, when $k > M$ and M is very large.

By definition, we write:

$$\langle r_i^+ \rangle_G \stackrel{\text{def}}{=} \langle \tilde{S}_{i+1} \rangle_G - \langle \tilde{S}_i \rangle_G, \quad (\text{A. 2a})$$

$$\langle r_i^- \rangle_G \stackrel{\text{def}}{=} \langle S_i \rangle_G - \langle \tilde{S}_i \rangle_G. \quad (\text{A. 2b})$$

Hypothesis (H. 5) and (A. 1) imply that

$$\langle \tilde{S}_i \rangle_G = \langle S_i \rangle_G + \langle \tilde{S}_{i+1} - \tilde{S}_i \rangle_G \sum_{k=1}^{M \rightarrow \infty} \frac{p_{i+k}}{p_i}, i \geq 1. \quad (\text{A. 3a})$$

Alternatively the same,

$$\langle r_i^- \rangle_G = -\langle r_i^+ \rangle_G \sum_{k=1}^{M \rightarrow \infty} \frac{p_{i+k}}{p_i}, \quad i \geq 1. \quad (\text{A. 3b})$$

By using the breakeven Hypothesis (H. 3) in (A. 3a), we write:

$$\langle r_i^+ \rangle_G = \langle \tilde{S}_{i+1} - \tilde{S}_i \rangle_G = \frac{p_i}{\sum_{i+1}} \left(\langle \tilde{S}_i \rangle_G - \frac{1}{i} \sum_{k=1}^i \langle \tilde{S}_k \rangle_G \right), \quad i > 1. \quad (\text{A. 4a})$$

Here, we denoted $\sum_i \stackrel{\text{def}}{=} \sum_{k=0}^{M \rightarrow \infty} p_{i+k} = p([N^*] \geq i)$.

Equation (A. 4a) is also:

$$\langle r_i^+ \rangle_G = \frac{p_i}{\sum_{i+1}} \frac{(i-1)}{i} \left(\langle \tilde{S}_i \rangle_G - \frac{1}{i-1} \sum_{k=1}^{i-1} \langle \tilde{S}_k \rangle_G \right), \quad i > 1. \quad (\text{A. 4b})$$

By applying (A. 4a) in (A. 4b), we get:

$$\langle r_i^+ \rangle_G = \frac{(i-1)}{i} \frac{p_i}{p_{i-1}} \frac{\sum_{i-1}}{\sum_{i+1}} \langle r_{i-1}^+ \rangle_G, \quad i > 2 \quad (\text{A. 5a})$$

and from (A.4a):

$$\langle r_2^+ \rangle_G = \frac{1}{2} \frac{p_2}{\sum_3} \langle r_1^+ \rangle_G. \quad (\text{A.5b})$$

Moreover, we could set:

$$\langle r_1^+ \rangle_G = \varepsilon \langle r_0^+ \rangle_G, \quad (\text{A.5c})$$

where ε is a constant, and $\varepsilon \neq p_1/\sum_2, 0$, to avoid the contradictions $\langle S_1 \rangle_G = \langle S_0 \rangle_G$ and $\langle r_i^+ \rangle_G = 0, \forall i \geq 1$, with $\pi \neq 0$.

In addition, equations (A.5) are reduced by recursion to:

$$\langle r_i^+ \rangle_G = \frac{\varepsilon}{i} p_i \frac{\sum_2}{\sum_{i+1}} \langle r_0^+ \rangle_G, \quad i \geq 2. \quad (\text{A.6})$$

Introducing (H.3) in the sequence $(i+1)\langle S_{i+1} \rangle_G - i\langle S_i \rangle_G$, and using (A.2b) we get

$$\langle S_{i+1} \rangle_G = \langle S_i \rangle_G - \frac{1}{i} \langle r_{i+1}^- \rangle_G, \quad i \geq 2. \quad (\text{A.7})$$

Matching (A.7) with (A.2b), we obtain

$$\langle \tilde{S}_{i+1} + r_{i+1}^- \rangle_G = \langle S_i \rangle_G - \frac{1}{i} \langle r_{i+1}^- \rangle_G. \quad (\text{A.8a})$$

From (A.8a), we arrive at the equation that establishes the relation between the market impact and the reversion (or permanent impact):

$$\langle \tilde{S}_{i+1} - S_i \rangle_G = -\frac{(i+1)}{i} \langle r_{i+1}^- \rangle_G, \quad i \geq 2. \quad (\text{A.8b})$$

This, using (A.3b) and (A.6), can be written as

$$\langle \tilde{S}_{i+1} - S_i \rangle_G = \frac{\varepsilon}{i} \frac{\sum_2}{\sum_{i+1}} \langle r_0^+ \rangle_G, \quad i \geq 2. \quad (\text{A.9})$$

Additionally, $\sum_2 = \sum_{k=0}^{M \rightarrow \infty} p_{2+k} = 1 - p_1$, and changing i per $j-1$, we get:

$$\langle \tilde{S}_j - S_{j-1} \rangle_G = \frac{\varepsilon(1-p_1)}{(j-1) \sum_{k=j}^{M \rightarrow \infty} p_k} \langle r_0^+ \rangle_G, \quad j \geq 3. \quad (\text{A.10a})$$

Finally, from (A.5c), (A.2b) and (A.3b), we obtain the interval impact for $j=2$, as:

$$\langle \tilde{S}_2 - S_1 \rangle_G = \frac{\varepsilon}{p_1} \langle r_0^+ \rangle_G. \quad (\text{A.10b})$$

Here, $\varepsilon \neq p_1/(1-p_1), 0, p_1$.

The empirical data show (Gopikrishnan *et al.*, 2000), (Gabaix *et al.*, 2006) that the hidden order sizes have a Pareto distribution with parameter $\alpha \in [1.5, 1.6]$:

$$p_k = \frac{1}{\zeta(\alpha + 1, 1) k^{\alpha+1}}, \quad k \geq 1. \quad (\text{A.11})$$

In this case, $\sum_{k=j}^{M \rightarrow \infty} p_k = p_1 \zeta(\alpha + 1, j)$, where $\zeta(\alpha + 1, j)$ is the Hurwitz Zeta function.

Formula (A.10b) should be consistent with (H.4), therefore it should be proportional to powers of the total transactions ξ_2^C , and the participation rate π_2 , which in this case are $\pi_2 = \pi$, $\xi_2^C = 2\pi^{-1}$. We know that $r_0^+ = \hat{\mu}(\sigma)\pi_1^\gamma$, and setting $\beta \stackrel{\text{def}}{=} \gamma + \alpha - 1$, and $\mu(\sigma) \stackrel{\text{def}}{=} \left(\frac{\varepsilon}{p_1}\right)(1 - p_1)\alpha \hat{\mu}(\sigma)$, we obtain:

$$\langle \tilde{S}_2 - S_1 \rangle_G = \frac{\mu(\sigma)}{2^{(\alpha-1)}(1 - p_1)\alpha} \pi^\beta \xi_2^{C(\alpha-1)}. \quad (\text{A.10c})$$

Formula (A.10c) enables us to make the connection with the impact at k for a variable participation schedule, when $\xi_2^C = \xi_k$, $\pi = \pi_k$, or the same $j = 2 = \xi_k \pi_k$. This is:

$$\langle \tilde{S}_k - S_{k-1} \rangle_G = \frac{\mu(\sigma)}{2^{(\alpha-1)}(1 - p_1)\alpha} \pi_k^\beta \xi_k^{(\alpha-1)}.$$

We re-obtained equation (4.2) when $k \neq 1$, and $\xi_k = 2\pi_k^{-1}$.

Appendix B. Derivation of Market Prices as Functions of the Participation Rate Schedule

The series of equations (7) may be proven by complete (or strong) induction. We will show that equation (7) is valid for $k = 1, 2$, and then is valid for $k = 3$.

First, note that we may write:

$$\tilde{S}_k = \langle \tilde{S}_k \rangle_G + \sum_{i=1}^k \sigma \tau_i^{\frac{1}{2}} \varsigma_i, \quad \forall k \geq 1, \quad (\text{B.1})$$

where

$$\langle \tilde{S}_k \rangle_G \stackrel{\text{def}}{=} S_0 + \text{Impact}_k + \sum_{i=1}^{k-1} PI_i. \quad (\text{B.2})$$

Here, PI_i is the notation for the permanent impact in each i -interval, and Impact_k is the sum of the permanent and temporary impacts at the k -interval as: $\text{Impact}_k = PI_k + TI_k$.

Demonstration of equations (B):

We write, following definitions H.2.1 and the equation hypothesis (h.2.4), that:

$$\begin{aligned}\tilde{S}_k &= S_0 + \sum_{i=1}^k (\tilde{S}_i - \tilde{S}_{i-1}) = S_0 + \sum_{i=1}^k (\tilde{S}_i - S_{i-1}) - \sum_{i=1}^{k-1} T I_i = S_0 + \\ &\left\{ \sum_{i=1}^k \text{Impact}_i + \sum_{i=1}^k \sigma \tau_i^{\frac{1}{2}} \varsigma_i \right\} - \sum_{i=1}^{k-1} T I_i = S_0 + \text{Impact}_k + \\ &\left\{ \sum_{i=1}^{k-1} (P I_i + T I_i) - \sum_{i=1}^{k-1} T I_i \right\} + \sum_{i=1}^k \sigma \tau_i^{\frac{1}{2}} \varsigma_i. \quad QED\end{aligned}$$

Proof of equations (7):

For $k = 1$, equation (7.2) follows from (B) and formula (4.2).

For $k = 2$, we have from (B. 1), (h. 2.4), and (4.2) that:

$$\tilde{S}_2 = (\langle S_1 \rangle_G + \mu \pi_2^\beta \xi_2^{\alpha-1}) + \sigma \sum_{i=1}^2 \pi_i^{-1} \varsigma_i.$$

The breakeven hypothesis (H.3) is valid for $k \geq 2$, but we may extend it to $k = 1$ taking¹³ $|T I_1 / \text{Impact}_1| \ll 1$ so $\langle S_1 \rangle_G \sim \langle \tilde{S}_1 \rangle_G$.

Then, after applying (7.2), $\tilde{S}_2$ becomes:

$$\tilde{S}_2 = S_0 + \mu \left\{ \pi_2^\beta \xi_2^{\alpha-1} + \pi_1^{\beta-\alpha+1} \right\} + \sigma \sum_{i=1}^2 \pi_i^{-1} \varsigma_i. \quad QED.$$

Finally, we apply the same reasoning to $\tilde{S}_3$ and we get:

$$\begin{aligned}\tilde{S}_3 &= \left\{ \xi_2^{-1} \pi_1^{-1} (S_0 + \mu \pi_1^{\beta-\alpha+1}) + \xi_2^{-1} \pi_2^{-1} (S_0 + \mu \pi_1^{\beta-\alpha+1} \right. \\ &\quad \left. + \mu \pi_2^\beta \xi_2^{\alpha-1}) \right\} + \mu \pi_3^\beta \xi_3^{\alpha-1} + \sigma \sum_{i=1}^3 \pi_i^{-1} \varsigma_i,\end{aligned}$$

that after reordering terms becomes:

$$\tilde{S}_3 = S_0 + \mu \left\{ \pi_3^\beta \xi_3^{\alpha-1} + \sum_{i=1}^2 \pi_i^{\beta-1} \xi_i^{\alpha-2} \right\} + \sigma \sum_{i=1}^3 \pi_i^{-1} \varsigma_i. \quad QED.$$

Therefore, formula (7) is valid for all $k \geq 1$ by strong induction.

Comments:

➤ Comparing formulas (7) with (B. 2), we get the expressions for temporary and permanent impacts in the approximation $k > 1$, as:

$$\begin{aligned}P I_k &= \mu(\sigma) \pi_k^{\beta-1} \xi_k^{\alpha-2}, \\ T I_k &= \mu(\sigma) \pi_k^{\beta-1} \xi_k^{\alpha-2} (\pi_k \xi_k - 1).\end{aligned}$$

➤ If instead of approximation $\langle S_1 \rangle_G \sim \langle \tilde{S}_1 \rangle_G$ we take $\langle S_1 \rangle_G = \langle \tilde{S}_1 \rangle_G - T I_1$, formulas (7) will become:

$$\langle \tilde{S}_2 \rangle_G = S_0 + \mu \left\{ \pi_2^\beta \xi_2^{\alpha-1} + \pi_1^{\beta-\alpha+1} \right\} - T I_1,$$

¹³ In the first interval, the hidden order has not yet been detected therefore expectations of continuing order flow imbalances are small. This makes the difference $\langle \tilde{S}_1 \rangle_G - \langle S_1 \rangle_G \stackrel{\text{def}}{=} T I_1 \rightarrow 0$ and $T I_1 \ll P I_1$, which means that market impact in the first interval consists predominantly of permanent impact.

$$\langle \tilde{S}_k \rangle_G = S_0 + \mu \left\{ \pi_k^\beta \xi_k^{\alpha-1} + \sum_{i=1}^{k-1} \pi_i^{\beta-1} \xi_i^{\alpha-2} \right\} - \frac{1}{(1+\pi_2\pi_1^{-1})} T I_1.$$

Moreover, for formula (9), the correction $\Delta \langle E \rangle_G$ to the cost will be:

$\Delta \langle E \rangle_G = n \xi_{N^*} \frac{\pi_2^{-1}}{\xi_2} T I_1$, with $T I_1 \rightarrow 0$. Because n, ξ_{N^*} are constants and $T I_1$ is only a function on π_1 , the correction term is only a function on π_1 and π_2 .

Appendix C. Derivation of the Cost of the Trade and its Variance

We will arrive to equations (9) and (10), by using the notations $n_i = n \pi_i^{-1}$, $\xi_k \stackrel{\text{def}}{=} \sum_{i=1}^k \frac{1}{\pi_i}$, and $\xi_{N^*} \stackrel{\text{def}}{=} (\xi_{[N^*]} + q \pi_f^{-1})$. From definition (8), we have:

$$\langle E(\vec{\pi}, N^*) \rangle_G \stackrel{\text{def}}{=} \langle n \xi_{N^*} S_0 - \sum_{k=1}^{[N^*]} n \pi_k^{-1} \tilde{S}_k - q n \pi_f^{-1} \tilde{S}_{N^*} \rangle_G.$$

By using formulas (7), this may be written as:

$$\langle E(\vec{\pi}, N^*) \rangle_G \stackrel{\text{def}}{=} T(S_0) + T(I),$$

with $T(S_0)$ containing the S_0 contributions and $T(I) = T(I, 1) + T(I, 2)$ the total impact, as:

$$T(S_0) = \left\{ n \xi_{N^*} S_0 - n S_0 \left(\sum_{k=1}^{[N^*]} \pi_k^{-1} + q \pi_f^{-1} \right) \right\} = 0,$$

$$T(I, 1) = -n \mu \left\{ \sum_{k=1}^{[N^*]} \pi_k^{\beta-1} \xi_k^{\alpha-1} + \sum_{k=2}^{[N^*]} \pi_k^{-1} \sum_{i=1}^{k-1} \pi_i^{\beta-1} \xi_i^{\alpha-2} \right\},$$

$$T(I, 2) = -n \mu q \pi_f^{-1} \left\{ \pi_f^\beta \xi_{N^*}^{\alpha-1} + \sum_{i=1}^{[N^*]} \pi_i^{\beta-1} \xi_i^{\alpha-2} \right\}.$$

The second summand in $T(I, 1)$ consists of two sums that may be switched as:

$$\sum_{k=2}^{[N^*]} \pi_k^{-1} \sum_{i=1}^{k-1} \pi_i^{\beta-1} \xi_i^{\alpha-2} = \sum_{i=1}^{[N^*]-1} \pi_i^{\beta-1} \xi_i^{\alpha-2} \sum_{k=i+1}^{[N^*]} \pi_k^{-1} = \sum_{i=1}^{[N^*]} \pi_i^{\beta-1} \xi_i^{\alpha-2} (\xi_{[N^*]} - \xi_i),$$

then:

$$T(I, 1) = -n \mu \xi_{[N^*]} \left\{ \sum_{k=1}^{[N^*]} \pi_k^{\beta-1} \xi_k^{\alpha-2} \right\}.$$

Summarizing, $\langle E(\vec{\pi}, N^*) \rangle_G = T(I, 1) + T(I, 2)$ as:

$$\langle E(\vec{\pi}, N^*) \rangle_G = -n\mu\xi_{N^*} \left\{ q\pi_f^{\beta-1}\xi_{N^*}^{\alpha-2} + \sum_{k=1}^{[N^*]} \pi_k^{\beta-1}\xi_k^{\alpha-2} \right\}.$$

On the other hand, by using definition (8) and formula (B. 1), we can write:

$$E = \langle E \rangle_G + Noise,$$

where

$$Noise = -n\sigma \left\{ \sum_{k=1}^{[N^*]} \pi_k^{-1} (q\pi_f^{-1}\zeta_k + \sum_{i=1}^k \pi_i^{-1}\zeta_i) \right\}.$$

The second summand in the noise term contains a double sum that can be switched as:

$$\sum_{k=1}^{[N^*]} \pi_k^{-1} \sum_{i=1}^k \pi_i^{-1} \zeta_i = \sum_{k=1}^{[N^*]} \pi_k^{-1} \zeta_k \sum_{i=k}^{[N^*]} \pi_i^{-1} = \sum_{k=1}^{[N^*]} \pi_k^{-1} \zeta_k (\xi_{[N^*]} - \xi_{k-1}).$$

Then, it results:

$$Noise = -n\sigma \sum_{k=1}^{[N^*]} \pi_k^{-1} \zeta_k (\xi_{N^*} - \xi_{k-1}).$$

After that, it is simple to calculate the variance of the cost $V(\vec{\pi}, N^*) \stackrel{\text{def}}{=} \langle E^2 \rangle_G - \langle E \rangle_G^2$. Explicitly, it is:

$$V(\vec{\pi}, N^*) = \langle Noise^2 \rangle_G + 2 \langle Noise \rangle_G \langle E \rangle_G$$

By using $\langle \zeta_k \rangle_G = 0$, $\forall k$, and Markov correlations $\langle \zeta_k \zeta_j \rangle_G = \delta_{kj}$, then $\langle Noise \rangle_G = 0$ and

$$V(\vec{\pi}, N^*) = \langle Noise^2 \rangle_G = \sigma^2 n^2 \sum_{k=1}^{[N^*]} \pi_k^{-2} (\xi_{N^*} - \xi_{k-1})^2.$$